

AVERAGES OF DIAGONAL ELLIOTT-HALBERSTAM PROBLEM TWISTED BY MÖBIUS FUNCTION WITH SOBOLEV AND HÖLDER-ZYGMUND WEIGHTS

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ABSTRACT. Recalling that the so-called Elliott-Halberstam conjecture twisted by the Möbius function $\mu(n)$ claims that

$$\sum_{q \leq N^\theta} \max_{y \leq N} \max_{(a,q)=1} \left| \sum_{\substack{n \leq y \\ n \equiv a \pmod q}} \Lambda(n) \mu(N-n) - \frac{1}{\varphi(q)} \sum_{n \leq y} \Lambda(n) \mu(N-n) \right| \ll \frac{N}{\log(N)^A}$$

for every $A > 0$, where $0 < \theta < 1$ is fixed, and also recalling that the validity of this conjecture, in combination with the validity of the classical Elliott-Halberstam for suitable θ , proves the binary Goldbach conjecture, in this paper we study weighted average variants of this problem. We will show that, under Generalized Riemann Hypothesis, a weak version of the Gonek-Hejhal conjecture and working with weights belonging to the Sobolev space $W^{2,1}$ or in the Hölder-Zygmund spaces $\mathcal{C}^\delta$ for suitable range of δ , the bound of the average is consistent with the bound of the “diagonal versions” of this conjecture (that is, taking $y = N$ and taking $n \equiv N \pmod q$). In particular, in the case of weights in Sobolev space, the consistent upper bound holds for the whole $0 < \theta < 1$ and, in the case of weights in the Hölder-Zygmund class $\mathcal{C}^\delta$, for θ that depends on the choice of δ but still not below the $1/2 - 2\varepsilon$ threshold.

1. INTRODUCTION

1.1. Context. The binary Goldbach conjecture is one of the oldest problems in mathematics and it states that every even integer greater than 2 is the sum of two primes. Despite its elementary formulation, it has proven incredibly difficult to attack, due to its deep connections to some of the most complex problems in analytic number theory. Consequently, many strategies have been proposed to study this problem; a nice survey of these ideas can be found [19].

However, these attempts have proven to be completely unsuitable for a definitive answer to the main conjecture and it is, in fact, a common opinion that there are still serious obstacles to overcome that current tools seem not to be able to handle. A widely held view is that the main obstruction is the so-called parity problem. Roughly speaking, sieve methods are extremely effective at controlling the distribution of almost-primes, but they do not by themselves distinguish reliably between integers with an even number of prime factors and integers with an odd number of prime factors. This limitation is a serious obstacle in problems where the exact multiplicative structure matters (for a more detailed analysis, see [10], chapt. 16). A classical illustration is provided by Chen’s theorem [6], which states that every sufficiently large even integer is the sum of a prime and an integer with at most two prime factors; the theorem comes remarkably close to Goldbach’s problem, but the remaining gap reflects precisely the parity barrier. A similar theorem to Chen’s result can be found in [6] regarding the twin prime

2020 *Mathematics Subject Classification.* 11N37, 11P32, 11N56.

Key words and phrases. Goldbach’s problem, Elliott-Halberstam conjecture, explicit formulae, weighted averages.

conjecture with the same limitations; this is not surprising, since these two conjectures are widely believed to be closely connected, and one may reasonably expect that a strategy proving one of them would also shed light on the other.

The parity problem is closely related to the behavior of oscillatory multiplicative functions, especially the Möbius function

$$\mu(n) := \begin{cases} 1, & n = 1 \\ (-1)^k, & n = p_1 \cdots p_k, k \in \mathbb{N}, p_j \text{ distinct prime numbers} \\ 0, & \text{otherwise.} \end{cases}$$

A useful heuristic principle, often called the Möbius randomness principle, is that $\mu(n)$ should exhibit substantial cancellation when correlated with sufficiently structured sequences. In informal terms, for a “reasonable” complex sequence a_n , the twisted sum $\sum_{n \leq x} \mu(n) a_n$ is expected to be much smaller than its trivial bound because of cancellation (see [16], chap. 13, for more details).

Many attempts have been made to overcome the parity barrier, see e.g. [9, 25]. In 2017 Murty and Vatswani [21] postulated a new conjecture that breaks down such obstacle, connecting the Elliott-Halberstam conjecture and the equidistribution of the Möbius function on shifted primes. We begin by recalling the classical Elliott-Halberstam conjecture.

Conjecture 1. (*Elliott-Halberstam conjecture* $EH(N^\theta \log(N)^C)$) *Let $0 < \theta < 1$ and $C > 0$ be fixed. Then, for every $A > 0$ we have*

$$\sum_{q \leq N^\theta \log(N)^C} \max_{y \leq N} \max_{(a,q)=1} \left| \sum_{\substack{n \leq y \\ n \equiv a \pmod q}} \Lambda(n) - \frac{y}{\varphi(q)} \right| \ll_A \frac{N}{\log(N)^A}$$

for every sufficiently large natural number N , where $\Lambda(n)$ is the Von Mangoldt function and $\varphi(n)$ is the Euler totient function.

Clearly, if $\theta < 1/2$, it is, up to the usual technical normalization of the level of distribution, the well-known Bombieri-Vinogradov theorem (see, e.g., [16], chap. 17). In that sense, Bombieri-Vinogradov may be viewed as a “satisfactory substitute”, citing [16], of the GRH, which is, today, out of reach. Murty and Vatswani proposed a version of the conjecture that connect EH and the distribution of the Möbius function.

Conjecture 2. (*Shifted Möbius Elliott-Halberstam Conjecture* $EH_{\mu_h}(N^\theta)$) *Let $0 < \theta < 1$ be fixed. Then, for any $A > 0$, we have*

$$\sum_{q \leq N^\theta} \max_{y < N} \max_{(a,q)=1} \left| \sum_{\substack{n \leq y \\ n \equiv a \pmod q}} \Lambda(n) \mu(n+h) - \frac{1}{\varphi(q)} \sum_{n \leq y} \Lambda(n) \mu(n+h) \right| \ll_A \frac{N}{\log(N)^A}$$

for every sufficiently large natural number N .

One of the main results in [21] is that a suitable combination of the classical Elliott-Halberstam conjecture and its shifted Möbius-twisted analogue would break the parity barrier in a strong quantitative form for the twin primes problem. Note that, in what follows, p will always denote a prime number.

Theorem 3. (Theorem 1.1 of [21]) Let $h \neq 0$ be a fixed even integer. Suppose that the conjectures $EH(N^\theta \log(N)^C)$ and $EH_{\mu_h}(N^{1-\theta})$ are true for some fixed $\theta < 1$ and a suitably large fixed C . Then, for all sufficiently large positive integers N , we then obtain the following:

1) We have

$$\sum_{n \leq N} \Lambda(n) \Lambda(n+h) \geq (1+o(1)) \mathfrak{S}_t(h) (1 - \mathcal{A}_h) N$$

where

$$\mathcal{A}_h := \prod_{\substack{p \nmid h \\ p > 2}} \left(1 - \frac{1}{p(p-1)}\right)$$

and

$$\mathfrak{S}_t(h) := \prod_{p|h} \left(1 + \frac{1}{p-1}\right) \prod_{p \nmid h} \left(1 - \frac{1}{(p-1)^2}\right)$$

is the singular series for the twin primes conjecture.

2) The asymptotic formula

$$\sum_{n \leq N} \Lambda(n) \Lambda(n+h) \sim \mathfrak{S}_t(h) N$$

is equivalent to the condition

$$\sum_{n \leq N} \Lambda(n) \mu(n+h) = o(N).$$

Note that, by the Bombieri-Vinogradov theorem, the previous result implies that the proof of the twin prime conjecture follows from the proof of the $EH_{\mu_h}(N^\theta)$ conjecture for $\theta > 1/2$.

The same ideas apply to Goldbach's problem. Indeed, Huang and Li proposed the following twisted version with the Möbius version of the EH conjecture (see [13]).

Conjecture 4. (Elliott-Halberstam conjecture twisted by Möbius function $EH_\mu(N^\theta)$) Let $0 < \theta < 1$ be fixed. Then, for every $A > 0$ we have

$$\sum_{q \leq N^\theta} \max_{y < N} \max_{(a,q)=1} \left| \sum_{\substack{n \leq y \\ n \equiv a \pmod q}} \Lambda(n) \mu(N-n) - \frac{1}{\varphi(q)} \sum_{n \leq y} \Lambda(n) \mu(N-n) \right| \ll_A \frac{N}{\log(N)^A}$$

for every sufficiently large natural number N .

From these conjectures, the mentioned authors proved the following theorem (see [13]).

Theorem 5. For a fixed $A > 0$ assume that $EH(N^\theta \log(N)^{2A+8})$ and $EH_\mu(N^{1-\theta})$ hold for some $0 < \theta < 1$. Then, for all sufficiently large positive integers N , the following hold:

1) We have

$$R(N) := \sum_{n < N} \Lambda(n) \Lambda(N-n) \geq N \mathfrak{S}(N) (1 - \mathcal{A}_N) + O_{A,\theta} \left(\frac{N}{\log(N)^A} \right)$$

where

$$\mathfrak{S}(N) := \begin{cases} 2 \prod_{\substack{p|N \\ p > 2}} \left(1 + \frac{1}{p-2}\right) \prod_{p > 2} \left(1 - \frac{1}{(p-1)^2}\right), & N \text{ even} \\ 0, & N \text{ odd} \end{cases}$$

is the singular series for the binary Goldbach conjecture and $\mathcal{A}_N$ is the same of Theorem 3.

2) The assertions

$$R(N) \sim N\mathfrak{S}(N)$$

and

$$\sum_{n < N} \Lambda(n) \mu(N - n) = o(N)$$

are equivalent.

As above, once the classical Elliott-Halberstam input is supplied by Bombieri-Vinogradov, proving $\text{EH}_\mu(N^\theta)$ for some $\theta > 1/2$ would yield the binary Goldbach conjecture for all sufficiently large even integers. The work of Huang and Li may also be viewed as a continuation of earlier work of Pan [24], who identified a related obstruction through the analysis of a truncated remainder term of the form

$$R^*(N) := \sum_{n < N} \left(\sum_{\substack{d_1 | n \\ d_1 > Q}} \mu(d_1) \log(d_1) \right) \left(\sum_{\substack{d_2 | (N-n) \\ (d_2, N)=1 \\ d_2 > Q}} \mu(d_2) \log(d_2) \right)$$

where the truncation at $Q := N^{1/2} (\log(N))^{-20}$ depends on the Bombieri-Vinogradov theorem.

1.2. Aims of the paper and methods of proof. The main goals of this paper are to study weighted average variants of $\text{EH}_\mu(N^\theta)$, that is

$$(1.1) \quad \sum_n \sum_m \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right), \sum_n \sum_m \Lambda(n) \chi(n) \mu(m) \log(m) f\left(\frac{n+m}{N}\right)$$

where $N \geq 4$ is a natural number, f is a suitable weight and χ is a primitive non-principal Dirichlet character mod q , where $q > 1$ is an integer. These quantities are natural because, using orthogonality of Dirichlet characters and making simple considerations, one allows to rewrite congruence sums, always assuming $(a, q) = 1$ and $y < N$, such as

$$\begin{aligned} & \sum_{\substack{n \leq y \\ n \equiv a \pmod{q}}} \Lambda(n) \mu(N - n) - \frac{1}{\varphi(q)} \sum_{n \leq y} \Lambda(n) \mu(N - n) \\ &= \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \bar{\chi}(a) \sum_{n \leq y} \Lambda(n) \chi^*(n) \mu(N - n) + O(\omega(q) \log(N)) \end{aligned}$$

where $\omega(q) := \sum_{p|q} 1 \ll \log(q)$ and χ^* is the primitive character that induces χ , so the average (1.1) is what we need, paying a reasonable error. We actually consider variants of $\text{EH}_\mu(N^\theta)$ in order to restore a useful symmetry between the prime variable and the Möbius variable: unlike the original form of $\text{EH}_\mu(N^\theta)$, where the summation is cut off at y but the twist is evaluated at $N - n$, the weighted expression depends symmetrically on the combination $n + m$. This symmetry is an important feature of our method. Moreover, in [13] (but actually also in [21], since the proofs are very similar) it is crucial to have $\max_{y \leq N} \sum_{n \leq y}$ because it is used to derive a logarithmically weighted variant, namely a bound for the same discrepancy with $\mu(N - n) \log(N - n)$ in place of $\mu(N - n)$. Thus, one cannot simply discard the $\max_{y \leq N}$, unlike $\max_{(a, q)=1}$ that actually can

be removed (see [13, 21]), without replacing it by a different and suitably robust hypothesis. For this reason, we propose the following modified, and in some sense weaker, version of $\text{EH}_\mu(N^\theta)$.

Conjecture 6. (*Diagonal Elliott-Halberstam conjecture twisted by Möbius function and logarithm $d\text{EH}_{\mu,\log}(N^\theta)$*). Let $0 < \theta < 1$ be fixed. Then, for every $A > 0$ we have

$$\sum_{\substack{q \leq N^\theta \\ (N,q)=1}} \left| \sum_{\substack{n < N \\ n \equiv N \pmod{q}}} \Lambda(n) g(N-n) - \frac{1}{\varphi(q)} \sum_{n < N} \Lambda(n) g(N-n) \right| \ll_A \frac{N}{\log(N)^A}$$

for every sufficiently large natural number N , where $g(n) \equiv \mu(n)$ or $g(n) \equiv \mu(n) \log(n)$.

This conjecture removes the two maxima appearing in $\text{EH}_\mu(N^\theta)$, but in exchange it requires the same strength of cancellation both in the pure Möbius case and in the logarithmically weighted case.

However, in Section 2, we show that if we replace Conjecture 4 with Conjecture 6, then Theorem 5 still holds, and this fact justifies our study of (4.1). In the same section we also show that the maximum over the residue class can be removed from the classical Elliott-Halberstam input as well.

After that, we consider (4.1) with different types of weights: if f is sufficiently regular, that is, it has its support contained in $[\alpha, \beta)$ and its restriction $f|_{(\alpha, \beta)}$ belongs to the Sobolev space $W^{2,1}(\alpha, \beta)$, then we prove, under GRH and assuming the simplicity of the non-trivial zeros of the Riemann zeta function $\zeta(s)$, a truncated explicit formula for (1.1). In somewhat informal terms, the main tool for proving this explicit formula will be a version of Abel's summation formula in two dimensions that allows us to "decouple" the explicit formulas of the averages of the arithmetic functions that define the problem. This explicit formula implies an averaged estimate compatible with the expected order of magnitude in this averaged setting, of the shape

$$(1.2) \quad \sum_{\substack{1 < q \leq N^\theta \\ (N,q)=1}} \frac{1}{\varphi(q)} \left| \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{N\alpha < n \leq N\beta} \sum_{m \leq N\beta - n} \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right) \right| \\ \ll_\varepsilon N^{2-\varepsilon} E(f'')$$

where $E(f'')$ is an (explicit) error that depends only on f'' , for every $\theta < 1 - 2\varepsilon$ and where $\varepsilon > 0$ is a sufficiently small number. The same result holds in the logarithmically weighted case.

An important remark is that in (1.2), and during the rest of the paper, we commit the following abuse of notation for the sake of greater readability: the sum $\sum_{\chi \neq \chi_0}$ denotes the sum over all non-principal characters modulo q , while $\chi(n)$ is understood as the value of the primitive character inducing χ .

We also show that, taking a discrete version of the two dimensional Abel formula, we are able to work with different weights f . Essentially, the problem shifts from the analysis of integrals with second derivative of the weight to sums with forward differences of order 2 of the weight. In order to have a control over the growth of the forward differences of f , it becomes natural to consider functions that lie in appropriate Hölder-Zygmund spaces.

More precisely, we show that if f is in the Hölder-Zygmund class $\mathcal{C}^\delta(\mathbb{R})$, $1 \leq \delta < 2$, again with the support of f contained in $[\alpha, \beta)$, then we obtain a truncated explicit formula and a

corresponding averaged bound compatible with the expected conjectural size, that is

$$\sum_{\substack{1 < q \leq N^\theta \\ (N, q) = 1}} \frac{1}{\varphi(q)} \left| \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{N\alpha < n \leq N\beta} \sum_{m \leq N\beta - n} \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right) \right| \ll_{\varepsilon, \delta, \alpha, \beta} N^{2-\varepsilon}$$

for $\theta < \delta - 1 - 2\varepsilon$ if $\delta \in [\frac{3}{2}, 2)$ and for $\theta < \frac{1}{2} - 2\varepsilon$ if $\delta \in [1, 3/2)$. Finally, we apply the same strategy to the case with logarithmic weighting, which allow us to show, in both the continuous and discrete cases, results similar to the previous ones.

The paper is organized as follows. In Section 2, we show that the diagonal conjecture $\text{dEH}_{\mu, \log}$ is sufficient for the proof of the Goldbach's problem. Section 3 collects the preliminary definitions, lemmas, and auxiliary tools used later on. Section 4 contains the first main results: we prove the truncated explicit formulae and the upper bounds for the weighted average of $\text{dEH}_{\mu, \log}$ with $g(n) = \mu(n)$ in both cases $W^{2,1}$ and $\mathcal{C}^\delta$. In Section 5 we prove similar results for the logarithmic case of $\text{dEH}_{\mu, \log}$ with $g(n) = \mu(n) \log(n)$. In Section 6 we show that some interesting examples of averages are a special cases of our results, such as, for instance, the well-known Cesàro-Riesz averages.

1.3. Notation. We use $Y \ll Z$, $Y = O(Z)$ to denote the estimate $|Y| \leq CZ$ for some suitable $C > 0$. If the constant C depends on some parameters $A, B, C \dots$ we write $Y \ll_{A, B, C, \dots} Z$ or $Y = O_{A, B, C, \dots}(Z)$. We write $Y \asymp Z$ if $Y \ll Z \ll Y$.

We write $\sum_{\chi \neq \chi_0}$ with the meaning of the sum on all the Dirichlet characters $\pmod{q}$ excluding the principal one but, with an abuse of notation, $\chi(n)$ is always a primitive character that induces χ . We denote

$$\sum_{n \leq A} := \sum_{0 < n \leq A}$$

and in all other cases it will be explicitly written from which integer the sum starts.

With the symbol $C^n(A)$, $n \in \mathbb{N}$, where $A = \mathbb{R}$ or A is an interval, we indicate the space of continuous function whose derivatives up to the n -th order are continuous in A . Clearly, $C^0(A)$ is the space of the continuous functions in A . With $AC(A)$ we indicate the space of the absolutely continuous functions in A .

With $L^\infty(\mathbb{R})$ we indicate the space of measurable functions in $\mathbb{R}$ whose essential supremum is finite, equipped with the norm

$$\|f\|_\infty := \text{ess sup}_{x \in \mathbb{R}} |f(x)|.$$

With $\Gamma(z)$ we denote the Euler Gamma function

$$\Gamma(z) := \int_0^{+\infty} x^{z-1} e^{-x} dx, \text{Re}(z) > 0,$$

with $B(z, h)$ the Beta function

$$B(z, h) := \int_0^1 x^{z-1} (1-x)^{h-1} dx = \frac{\Gamma(z) \Gamma(h)}{\Gamma(z+h)}, \text{Re}(z) > 0, \text{Re}(h) > 0$$

and with $\psi^{(0)}(z)$ the Digamma function

$$\psi^{(0)}(z) := \frac{\Gamma'(z)}{\Gamma(z)}, z \in \mathbb{C}, z \neq 0, -1, -2, \dots$$

2. PROOF OF GOLDBACH'S CONJECTURE ASSUMING CONJECTURE 6

In this section we prove a version of Theorem 5 with Conjecture 6. We will often recall some results in [13] because the proof is, essentially, the same.

Theorem 7. *Let $0 < \theta < 1$ and $C > 0$ be fixed. Assume that*

$$(2.1) \quad \sum_{\substack{q \leq N^\theta \log(N)^C \\ (N, q) = 1}} \left| \sum_{\substack{n < N \\ n \equiv N \pmod{q}}} \Lambda(n) - \frac{N}{\varphi(q)} \right| \ll_A \frac{N}{\log(N)^A}$$

for every sufficiently large natural number N and $dEH_{\mu, \log}(N^{1-\theta})$ holds for the same θ . Then, using the same notations of Theorem 5, we have

$$R(N) := \sum_{n < N} \Lambda(n) \Lambda(N-n) \geq N \mathfrak{S}(N) (1 - \mathcal{A}_N) + O_{A, \theta} \left(\frac{N}{\log(N)^A} \right).$$

Moreover, the assertions $R(N) \sim N \mathfrak{S}(N)$ and $\sum_{n < N} \Lambda(n) \mu(N-n) = o(N)$ are equivalent.

Proof. Fix $0 < \theta < 1$ and let $\alpha = N^\theta$. Following [13], we write

$$R(N) = S_1(\alpha) + S_2(\alpha) + O\left(N^{1/2} \log(N)^3\right)$$

and

$$S_1(\alpha) := \sum_{n < N} \Lambda(n) \mu(N-n)^2 \sum_{\substack{d | (N-n) \\ d \leq \alpha}} \mu(d) \log\left(\frac{1}{d}\right)$$

$$S_2(\alpha) := \sum_{n < N} \Lambda(n) \mu(N-n)^2 \sum_{\substack{d | (N-n) \\ d > \alpha}} \mu(d) \log\left(\frac{1}{d}\right).$$

Let us study $S_1(\alpha)$. From the identity

$$\mu(N-n)^2 = \sum_{b^2 | (N-n)} \mu(b)$$

we have

$$S_1(\alpha) = \sum_{d \leq \alpha} \mu(d) \log\left(\frac{1}{d}\right) \sum_{\substack{b < \sqrt{N} \\ [b^2, d] < N}} \mu(b) \sum_{\substack{n < N \\ n \equiv N \pmod{[b^2, d]}}} \Lambda(n).$$

Following section 3.1 of [13] we, split the sum to $b > B := \log(N)^{A+4}$ and $b \leq B$. This lead to

$$S_1(\alpha) = N \sum_{d \leq \alpha} \frac{\mu(d) \log(1/d)}{\varphi(d)} \sum_{\substack{b \leq B \\ (bd, N) = 1}} \frac{\mu(b) \varphi((b, d))}{b \varphi(b)}$$

$$+ E_1(\alpha) + O_A \left(\frac{N}{\log(N)^A} \right),$$

where

$$E_1(\alpha) \ll \log(N) \sum_{z \leq \alpha B^2} \tau_3(z) \left| \sum_{\substack{n < N \\ n \equiv N \pmod{z}}} \Lambda(n) - \frac{N}{\varphi(z)} \right|$$

$$\ll N^{1/2} \log(N)^{3/2} \left(\sum_{z \leq \alpha B^2} \frac{\tau_3(z)^2}{z} \right)^{1/2} \left(\sum_{z \leq \alpha B^2} \left| \sum_{\substack{n < N \\ n \equiv N \pmod{z}}} \Lambda(n) - \frac{N}{\varphi(z)} \right| \right)^{1/2}$$

by Cauchy-Schwarz and from the trivial bound

$$\left| \sum_{\substack{n < N \\ n \equiv N \pmod{z}}} \Lambda(n) - \frac{N}{\varphi(z)} \right| \ll \frac{N \log(N)}{z},$$

where

$$\tau_3(z) := \sum_{abc=z} 1.$$

Since, by partial summation, we have

$$\sum_{z \leq \alpha B^2} \frac{\tau_3(z)^2}{z} \ll \log(N)^9$$

(see section 3.1 in [13], p. 338 for more details) it remains to evaluate

$$\sum_{z \leq \alpha B^2} \left| \sum_{\substack{n < N \\ n \equiv N \pmod{z}}} \Lambda(n) - \frac{N}{\varphi(z)} \right| \ll_A \frac{N}{\log(N)^{2A+12}}$$

by (2.1). So, from the standard estimate for $\mathfrak{S}(N)$, we get

$$S_1(\alpha) = N \mathfrak{S}(N) + O\left(\frac{N}{\log(N)^A}\right).$$

Continuing to follow [13], we split $S_2(\alpha)$ in two parts

$$S_2(\alpha) = S_3(\alpha) - S_4(\alpha) + O\left(\frac{N}{\log(N)^A}\right)$$

where

$$S_3(\alpha) := \sum_{\substack{k < (N-1)/\alpha \\ (k, N)=1}} \mu(k) \log(k) \sum_{\substack{n < N \\ n \equiv N \pmod{k}}} \Lambda(n) \mu(N-n)$$

and

$$S_4(\alpha) := \sum_{\substack{k < (N-1)/\alpha \\ (k, N)=1}} \mu(k) \sum_{\substack{n < N \\ n \equiv N \pmod{k}}} \Lambda(n) \mu(N-n) \log(N-n).$$

We analyze $S_3(\alpha)$. We have

$$S_3(\alpha) = \sum_{\substack{k < (N-1)/\alpha \\ (k,N)=1}} \frac{\mu(k) \log(k)}{\varphi(k)} \sum_{n < N} \Lambda(n) \mu(N-n) + E_3(\alpha)$$

where

$$E_3(\alpha) := \sum_{\substack{k < (N-1)/\alpha \\ (k,N)=1}} \mu(k) \log(k) \left(\sum_{\substack{n < N \\ n \equiv N \pmod k}} \Lambda(n) \mu(N-n) - \frac{1}{\varphi(k)} \sum_{n < N} \Lambda(n) \mu(N-n) \right)$$

(see [13], p. 340) hence by Conjecture 6 with $g(n) = \mu(n)$ we get

$$E_3(\alpha) \ll_A \frac{N}{\log(N)^A}$$

and from the standard estimate

$$\sum_{\substack{k < R \\ (k,N)=1}} \frac{\mu(k) \log(k)}{\varphi(k)} = -\mathfrak{S}(N) + O\left(e^{-C\sqrt{\log(R)}}\right)$$

we conclude that

$$S_3(\alpha) = -\mathfrak{S}(N) \sum_{n < N} \Lambda(n) \mu(N-n) + O_A\left(\frac{N}{\log(N)^A}\right)$$

(see [13], p. 341 for more details). With a similar argument we have that

$$S_4(\alpha) = \sum_{\substack{k < (N-1)/\alpha \\ (k,N)=1}} \frac{\mu(k)}{\varphi(k)} \sum_{n < N} \Lambda(n) \mu(N-n) \log(N-n) + E_4(\alpha)$$

where

$$E_4(\alpha) := \sum_{\substack{k < (N-1)/\alpha \\ (k,N)=1}} \mu(k) \left(\sum_{\substack{n < N \\ n \equiv N \pmod k}} \Lambda(n) \mu(N-n) \log(N-n) - \frac{1}{\varphi(k)} \sum_{n < N} \Lambda(n) \mu(N-n) \log(N-n) \right)$$

and again by Conjecture 6 with $g(n) = \mu(n) \log(n)$ we get

$$E_4(\alpha) \ll_A \frac{N}{\log(N)^A}.$$

It remains to observe that

$$\sum_{\substack{k < (N-1)/\alpha \\ (k,N)=1}} \frac{\mu(k)}{\varphi(k)} \ll e^{-C\sqrt{\log((N-1)/\alpha)}}$$

and

$$\sum_{n < N} \Lambda(n) \mu(N-n) \log(N-n) \ll N \log(N)^2$$

to get

$$S_4(\alpha) \ll_A \frac{N}{\log(N)^A}$$

(see [13], p. 341, 342 for more details). The rest of the proof is exactly the same of [13]. $\square$

Therefore, using the Conjecture 6 allows us to effectively estimate the error terms $E_1(\alpha)$, $E_3(\alpha)$ and $E_4(\alpha)$ and thus prove the binary Goldbach's conjecture in exactly the same way as done in [13].

3. PRELIMINARY RESULTS, DEFINITIONS AND MAIN TOOLS

In this section we have to recall several facts and prove some preliminary results. For this reason, we divide this section further subsections.

3.1. The main tools. One of the main tools used to show our theorems is an identity proved in [3]. It can be viewed as a two-dimensional Abel summation formula. Versions of two-dimensional Abel summation do appear in the literature (see, e.g. [17] or [1]), but the presented formulation makes explicit the link between a weighted average involving two arithmetic functions and the Laplace convolution of the corresponding unweighted averages taken separately. This point of view is particularly useful, since it allows us to combine explicit formulae for the individual averages, whenever available, and thereby derive an explicit formula for the original weighted average. To the best of our knowledge, Abel summation has not previously been formulated in exactly this way in the literature.

Theorem 8. *Let g_1, g_2 be arithmetical functions, $\eta \in \mathbb{R}^+$, $f : \mathbb{R} \rightarrow \mathbb{C}$, and assume that:*

- 1) *f has its support in $[\alpha, \beta)$, $0 \leq \alpha < \beta$, and $\beta \in \mathbb{R}$*
- 2) *$f \in C^1(\alpha, \beta)$.*
- 3) *$f' \in AC(\alpha, \beta)$.*

Then, if

$$G_j(x) := \begin{cases} \sum_{n \leq x} g_j(n), & x > 0 \\ 0, & \text{otherwise} \end{cases}$$

for $j = 1, 2$, we get

$$\begin{aligned} \sum_{\eta\alpha < n \leq \eta\beta} \sum_{m \leq \eta\beta - n} g_2(m) g_1(n) f\left(\frac{m+n}{\eta}\right) &= G_2(\eta\alpha) \int_{\alpha}^{\beta} G_1(\eta v - \eta\alpha) f'(v) dv \\ &+ \frac{1}{\eta} \int_{\alpha}^{\beta} f''(w) \int_{\eta\alpha}^{\eta w} G_2(s) G_1(\eta w - s) ds dw. \end{aligned}$$

Observe that, essentially, we require that $f|_{(\alpha, \beta)}$ belongs to the Sobolev spaces $W^{2,1}(\alpha, \beta)$. We recall the definition, just for completeness: if I is an open interval, then

$$W^{1,p}(I) := \left\{ f \in L^p(I) : \exists g \in L^p(I) : \int_I f h' = - \int_I g h, \forall h \in C_c^1(I) \right\}$$

where $C_c^1(I)$ are the function $C^1(I)$ with compact support in I , and

$$W^{m,p}(I) := \{ f \in W^{m-1,p}(I) : f' \in W^{m-1,p}(I) \}, m \in \mathbb{N}, m \geq 2$$

(see, e.g., [2], chapt. 8). We recall that is possible to extend the results of the previous theorem also in the case $\beta = +\infty$ (see [3]). Note that the hypotheses of the Theorem imply that $f(\alpha^+)$, $f'(\alpha^+)$ exists and are finite (but not necessary equal to 0) and $f(\beta^-) = f'(\beta^-) = 0$.

In order to work with more general weights, we propose now a discrete version of the previous theorem. Firstly, we recall the definition of r -th order forward difference, with $r \in \mathbb{N}^+$, as

$$\Delta_h^r(f, x) := \sum_{k=0}^r \binom{r}{k} (-1)^{r-k} f(x + hk), \quad x, h \in \mathbb{R},$$

$$\Delta_h(f, x) := \Delta_h^1(f, x)$$

and the property

$$(3.1) \quad \Delta_h^r = \Delta_h [\Delta_h^{r-1}]$$

(see. e.g., [8], chapt. 7).

Theorem 9. *Let g_1, g_2 be arithmetical functions, let $f : \mathbb{R} \rightarrow \mathbb{C}$ and assume that:*

1) *f has its support in $[\alpha, \beta)$, $0 \leq \alpha < \beta$.*

2) *$G_1(1) = G_1(0) = G_2(0) = 0$.*

Then, taking $\eta \in \mathbb{R}^+$ such that $\eta\beta \in \mathbb{N}^+$, $\eta\alpha \in \{0, 1\}$ we have

$$\sum_{\eta\alpha < n \leq \eta\beta} \sum_{m \leq \eta\beta - n} g_2(m) g_1(n) f\left(\frac{m+n}{\eta}\right) = \sum_{k=2}^{\eta\beta-1} \left(\sum_{u=0}^k G_1(u) G_2(k-u) \right) \Delta_{1/\eta}^2\left(f, \frac{k}{\eta}\right).$$

Proof. By classical summation by parts and since $f(\beta) = f\left(\beta + \frac{1}{\eta}\right) = 0$, because f has its support in $[\alpha, \beta)$, we have

$$\sum_{m \leq \eta\beta - n} g_2(m) f\left(\frac{m+n}{\eta}\right) = - \sum_{m \leq \eta\beta - n - 1} G_2(m) \left[f\left(\frac{m+n+1}{\eta}\right) - f\left(\frac{m+n}{\eta}\right) \right]$$

and so defining

$$C(n) := - \sum_{m \leq \eta\beta - n - 1} G_2(m) \left[f\left(\frac{m+n+1}{\eta}\right) - f\left(\frac{m+n}{\eta}\right) \right]$$

we have, again by partial summation

$$\begin{aligned} \sum_{\eta\alpha < n \leq \eta\beta} g_1(n) \sum_{m \leq \eta\beta - n} g_2(m) f\left(\frac{m+n}{\eta}\right) &= G_1(\eta\beta) C(\eta\beta) - G_1(\eta\alpha) C(\eta\alpha + 1) \\ &\quad - \sum_{\eta\alpha < n \leq \eta\beta - 1} G_1(n) (C(n+1) - C(n)). \end{aligned}$$

Now clearly

$$C(\eta\beta) = G_1(\eta\alpha) = 0$$

hence

$$\begin{aligned} \sum_{\eta\alpha < n \leq \eta\beta} g_1(n) \sum_{m \leq \eta\beta - n} g_2(m) f\left(\frac{m+n}{\eta}\right) &= - \sum_{\eta\alpha < n \leq \eta\beta - 1} G_1(n) (C(n+1) - C(n)) \\ &= \sum_{\eta\alpha < n \leq \eta\beta - 1} G_1(n) \sum_{m \leq \eta\beta - n - 2} G_2(m) \left[f\left(\frac{m+n+2}{\eta}\right) - f\left(\frac{m+n+1}{\eta}\right) \right] \\ &\quad - \sum_{\eta\alpha < n \leq \eta\beta - 1} G_1(n) \sum_{m \leq \eta\beta - n - 1} G_2(m) \left[f\left(\frac{m+n+1}{\eta}\right) - f\left(\frac{m+n}{\eta}\right) \right] \end{aligned}$$

$$= \sum_{\eta\alpha < n \leq \eta\beta-1} G_1(n) \sum_{m \leq \eta\beta-n-1} G_2(m) \Delta_{1/\eta}^2 \left(f, \frac{m+n}{\eta} \right)$$

then, taking $k = m + n$ and using condition 2), we obtain

$$\sum_{\eta\alpha < n \leq \eta\beta} g_1(n) \sum_{m \leq \eta\beta-n} g_2(m) f \left(\frac{m+n}{\eta} \right) = \sum_{\eta\alpha < k \leq \eta\beta-1} \Delta_{1/\eta}^2 \left(f, \frac{k}{\eta} \right) \sum_{u=0}^k (G_1(u) G_2(k-u))$$

as wanted. $\square$

The previous result can be proved in a more general settings, but paying the price of having an identity with a more complicated form.

Corollary 10. *Under the previous hypotheses, we have*

$$\begin{aligned} \sum_{\eta\alpha < n \leq \eta\beta} g_1(n) \sum_{m \leq \eta\beta-n} g_2(m) f \left(\frac{m+n}{\eta} \right) &= \sum_{\eta\alpha < k \leq \eta\beta-1} \Delta_{1/\eta}^2 \left(f, \frac{k}{\eta} \right) \int_0^k G_1(s) G_2(k-s) ds \\ &+ \sum_{\eta\alpha < k \leq \eta\beta-1} \Delta_{1/\eta}^2 \left(f, \frac{k}{\eta} \right) \sum_{u=0}^k G_1(u) g_2(k-u). \end{aligned}$$

Proof. It is enough to observe that

$$\int_0^k G_1(s) G_2(k-s) ds = \sum_{u=0}^{k-1} \int_u^{u+1} G_1(s) G_2(k-s) ds = \sum_{u=0}^{k-1} G_1(u) G_2(k-u-1)$$

hence

$$\begin{aligned} \sum_{u=0}^k G_1(u) G_2(k-u) &- \sum_{u=0}^{k-1} G_1(u) G_2(k-u-1) \\ &= \sum_{u=0}^{k-1} G_1(u) [G_2(k-u) - G_2(k-u-1)] \\ &= \sum_{u=0}^{k-1} G_1(u) g_2(k-u). \end{aligned}$$

$\square$

3.2. Hölder-Zygmund spaces. As mentioned in the introduction, we are interested in weights defined by functions belonging to suitable function spaces. In the previous subsection we considered the case in which the restriction $f|_{(\alpha,\beta)} \in W^{2,1}(\alpha,\beta)$. In this subsection, we turn to the second setting under consideration, namely Hölder-Zygmund spaces. We take this class because the previous theorems show that we need an effective way to control finite differences of the weight. We therefore recall the definition of these spaces and some of their basic properties. Everything will be formulated on $\mathbb{R}$, since our functions are defined on $\mathbb{R}$, even when their support is contained in an interval; this is important because the behavior near the boundary is different.

Definition 11. (Hölder-Zygmund space) Let $\delta > 0$. We define the Hölder-Zygmund space as

$$\mathcal{C}^\delta(\mathbb{R}) := \left\{ f \in L^\infty(\mathbb{R}) \cap C^0(\mathbb{R}) : \|f\|_{\mathcal{C}^\delta(\mathbb{R})} < +\infty \right\}$$

where

$$\|f\|_{\mathcal{C}^\delta(\mathbb{R})} := \|f\|_\infty + \sup_{x,h \in \mathbb{R}, h \neq 0} \frac{|\Delta_h^1(f, x)|}{h^\delta}$$

if $0 < \delta < 1$,

$$\|f\|_{\mathcal{C}^1(\mathbb{R})} := \|f\|_{\infty} + \sup_{x, h \in \mathbb{R}, h \neq 0} \frac{|\Delta_h^2(f, x)|}{h}$$

if $\delta = 1$. If $\delta > 1$, say $m < \delta \leq m + 1$, $m \in \mathbb{N}$, then we define the Hölder-Zygmund space as

$$\mathcal{C}^{\delta}(\mathbb{R}) := \left\{ f \in L^{\infty}(\mathbb{R}) \cap C^m(\mathbb{R}) : \|f\|_{\mathcal{C}^{\delta}(\mathbb{R})} < +\infty \right\}$$

where the norm is defined recursively as

$$\|f\|_{\mathcal{C}^{\delta}(\mathbb{R})} = \|f\|_{\mathcal{C}^{\delta-1}(\mathbb{R})} + \|f'\|_{\mathcal{C}^{\delta-1}(\mathbb{R})}.$$

(see, e.g, [26]) Note that $\mathcal{C}^1(\mathbb{R})$ is also called Zygmund spaces, and if $\delta \notin \mathbb{N}$, $\mathcal{C}^{\delta}(\mathbb{R})$ is equivalent to the classical Hölder spaces. Moreover, we have the following result.

Theorem 12. *If $f \in L^{\infty}(\mathbb{R}) \cap C^0(\mathbb{R})$ and $0 < \delta < n$, $n \in \mathbb{N}$, then $f \in \mathcal{C}^{\delta}(\mathbb{R}) \Leftrightarrow |\Delta_h^n(f, x)| \leq C|h|^{\delta}$ for all $x, h \in \mathbb{R}$.*

For a reference, see [18], Theorem 6.1. We also recall that the following useful relations for function in Zygmund and Hölder-Zygmund class about the definite difference $\Delta_h^1(f, x)$; indeed, we have

$$(3.2) \quad f \in \mathcal{C}^1(\mathbb{R}) \Rightarrow |\Delta_h^1(f, x)| \leq M \left| h \log \left(\frac{1}{h} \right) \right|, \quad M > 0$$

for all $x, h \in \mathbb{R}$, $h \neq 0$ (see [5], Prop. 2.3.7) and, if $\delta > 1$

$$f \in \mathcal{C}^{\delta}(\mathbb{R}) \Rightarrow |\Delta_h^1(f, x)| \leq M|h|, \quad M > 0$$

that is, f verifies the classical Lipschitz condition, for all $x, h \in \mathbb{R}$, since

$$\mathcal{C}^{\delta}(\mathbb{R}) \hookrightarrow \text{Lip}(\mathbb{R})$$

if $\delta > 1$, where

$$\text{Lip}(\mathbb{R}) := \left\{ f \in L^{\infty}(\mathbb{R}) : \|f\|_{\text{Lip}(\mathbb{R})} < +\infty \right\}$$

and

$$\|f\|_{\text{Lip}(\mathbb{R})} = \|f\|_{\infty} + \sup_{x, h \in \mathbb{R}, h \neq 0} \frac{|\Delta_h^1(f, x)|}{|h|}$$

(see again [26]).

3.3. Estimate for ratio of Gamma function $\Gamma(x)$ and other results. It is a standard fact that, when one considers a discrete convolution of functions admitting an explicit formula involving series over zeros, the corresponding explicit formula for the convolution is related, in one way or another, to a sum involving a convolution of those zeros. Very often, this leads to the need to control series involving the Euler beta function or ratios of gamma functions. This phenomenon also arises in our work, so it is convenient to record the following proposition.

Proposition 13. *Let $x, y \in \mathbb{R}$. Then*

$$\left| \frac{\Gamma\left(\frac{1}{2} + ix\right) \Gamma\left(\frac{1}{2} + iy\right)}{\Gamma(2 + i(x+y))} \right| \ll \begin{cases} \frac{e^{\pi(|x+y| - |x| - |y|)/2}}{|x+y|^{3/2}}, & |x+y| > 1 \\ e^{-\pi(|x| + |y|)/2}, & |x+y| \leq 1, \end{cases}$$

Proof. From the well-known relations

$$\left| \Gamma \left(\frac{1}{2} + ix \right) \right|^2 = \frac{\pi}{\cosh(\pi x)}, \quad |\Gamma(2 + i(x+y))|^2 = \left(1 + (x+y)^2\right) \frac{\pi |x+y|}{\sinh(\pi |x+y|)}$$

(see [23], 5.4.3 and 5.4.4), then

$$\left| \frac{\Gamma(\frac{1}{2} + ix) \Gamma(\frac{1}{2} + iy)}{\Gamma(2 + i(x+y))} \right|^2 = \pi \frac{\sinh(\pi |x+y|)}{\left(1 + (x+y)^2\right) |x+y| \cosh(\pi x) \cosh(\pi y)}.$$

Assume $|x+y| > 1$. Then, since

$$\sinh(\pi |s|) \leq \frac{e^{\pi |s|}}{2}, \quad \cosh(\pi s) \geq \frac{e^{\pi |s|}}{2}$$

valid for all $s \in \mathbb{R}$, we get

$$\left| \frac{\Gamma(\frac{1}{2} + ix) \Gamma(\frac{1}{2} + iy)}{\Gamma(2 + i(x+y))} \right| \ll \frac{e^{\pi(|x+y|-|x|-|y|)/2}}{|x+y|^{3/2}}.$$

If $|x+y| \leq 1$, since

$$\frac{\sinh(\pi |s|)}{\pi |s|} \ll 1$$

then we have

$$\left| \frac{\Gamma(\frac{1}{2} + ix) \Gamma(\frac{1}{2} + iy)}{\Gamma(2 + i(x+y))} \right| \ll e^{-\pi(|x|+|y|)/2}.$$

□

Convergence of double series on the non-trivial zeros is a delicate matter; for this reason, in [4], we prove a theorem about the convergence of a double series over the non-trivial zeros of the Riemann zeta function $\zeta(s)$ and a suitable function $f(\rho)$ that verifies some hypotheses. We need a similar results but now for the non-trivial zeros of $L(s, \chi)$. We first recall the following result.

Proposition 14. *Let $T \geq 4$ be a real number and $N(T)$ be the number of non-trivial zeros $\rho = \sigma + i\gamma$ of $\zeta(s)$ in the rectangle $0 < \sigma < 1, 0 < \gamma < T$. Moreover, let χ a primitive character mod q with $q \in \mathbb{N}^+, q > 1$ and let $N(T, \chi)$ be the number of non-trivial zeros $\rho_\chi = \sigma_\chi + i\gamma_\chi$ in the rectangle $0 < \sigma_\chi < 1, 0 \leq \gamma_\chi \leq T$. Then, we have*

$$(3.3) \quad N(T) = \frac{T}{2\pi} \log \left(\frac{T}{2\pi} \right) - \frac{T}{2\pi} + O(\log(T)),$$

$$(3.4) \quad N(T, \chi) = \frac{T}{2\pi} \log \left(\frac{qT}{2\pi} \right) - \frac{T}{2\pi} + O(\log(qT)).$$

See, e.g, [7], chapt. 15 and 16 or [20], chapt. 14. Clearly, from the previous formulae we may derive also the cases $|\gamma| \leq T, |\gamma_\chi| \leq T$, due to the symmetry of the zeros of $\zeta(s)$ and from the fact that the number of zeros of $L(s, \chi)$ with $-T \leq \gamma_\chi \leq 0$ is $N(T, \bar{\chi})$.

Theorem 15. *Assume the generalized Riemann hypothesis, fix $q_1, q_2 \in \mathbb{N}^+$ and $\chi_1 \pmod{q_1}, \chi_2 \pmod{q_2}$ two Dirichlet character. Let $\rho_{\chi_j} = \frac{1}{2} + i\gamma_{\chi_j}$ runs over the non-trivial zeros of the L -function $L(s, \chi_j)$, $j = 1, 2$. Let*

$$Z_0^+(\chi_j) := \{\rho_{\chi_j} : L(\rho_{\chi_j}, \chi_j) = 0, \gamma_{\chi_j} \geq 0\},$$

$$Z^+(\chi_j) := \{\rho_{\chi_j} : L(\rho_{\chi_j}, \chi_j) = 0, \gamma_{\chi_j} > 0\}$$

and assume that with the notation

$$\sum_{\rho_{\chi_j} \in Z_0^+(\chi_j)} , \quad \sum_{\rho_{\chi_j} \in Z^+(\chi_j)}$$

we intend that we are taking $\rho_{\chi_j} \in Z_0^+(\chi_j)$, $\rho_{\chi_j} \in Z^+(\chi_j)$ and summing it considering their multiplicity, for $j = 1, 2$. Let $f(z)$, $g(z)$ be complex functions defined on the whole set of the non-trivial zeros of $L(s, \chi_1)$, $L(s, \bar{\chi}_1)$ and $L(s, \chi_2)$, $L(s, \bar{\chi}_2)$, respectively. Moreover, assume

$$\begin{aligned} \sum_{\substack{\rho_{\chi_1} \in Z_0^+(\chi_1) \\ \gamma_{\chi_1} \leq T}} \left| \frac{f(\rho_{\chi_1})}{\rho_{\chi_1}} \right| &= o(T^\alpha), \quad \sum_{\substack{\rho_{\chi_1} \in Z_0^+(\chi_1) \\ \gamma_{\chi_1} \leq T}} \left| \frac{f(\overline{\rho_{\chi_1}})}{\rho_{\chi_1}} \right| = o(T^\alpha) \\ \sum_{\substack{\rho_{\chi_2} \in Z_0^+(\chi_2) \\ \gamma_{\chi_2} \leq T}} \left| \frac{g(\rho_{\chi_2})}{\rho_{\chi_2}} \right| &= o(T^\alpha), \quad \sum_{\substack{\rho_{\chi_2} \in Z_0^+(\chi_2) \\ \gamma_{\chi_2} \leq T}} \left| \frac{g(\overline{\rho_{\chi_2}})}{\rho_{\chi_2}} \right| = o(T^\alpha) \end{aligned}$$

as $T \rightarrow +\infty$ for every $\alpha > 0$. Then, the double series

$$\sum_{\rho_{\chi_1}} f(\rho_{\chi_1}) \sum_{\rho_{\chi_2}} g(\rho_{\chi_2}) \frac{\Gamma(\rho_{\chi_1}) \Gamma(\rho_{\chi_2})}{\Gamma(\rho_{\chi_1} + \rho_{\chi_2} + k + 1)}$$

converges absolutely for $k \in \mathbb{R}$, $k > 1/2$.

Proof. Fix $\chi_1 \pmod{q_1}$, $\chi_2 \pmod{q_2}$. We may assume that $\gamma_{\chi_j} \neq 0$, $j = 1, 2$ because, if one them is equal to zero, we have

$$\begin{aligned} \sum_{\rho_{\chi_1}} f(\rho_{\chi_1}) \sum_{\rho_{\chi_2}: \gamma_{\chi_2}=0} g(\rho_{\chi_2}) \frac{\Gamma(\rho_{\chi_1}) \Gamma(\rho_{\chi_2})}{\Gamma(\rho_{\chi_1} + \rho_{\chi_2} + k + 1)} \\ \ll_f N(2, \chi_2) \sum_{\rho_{\chi_1}} |f(\rho_{\chi_1})| \left| \frac{\Gamma(\rho_{\chi_1})}{\Gamma(\rho_{\chi_1} + k + \frac{3}{2})} \right| \end{aligned}$$

and the series, for $k > 1/2$, trivially converges (actually it converges for smaller k , but it is not important for our aims) by the Stirling's formula

$$(3.5) \quad |\Gamma(x + iy)| \sim e^{-\pi|y|/2} |y|^{x-1/2}, \quad x_1 < x < x_2, \quad |y| \rightarrow +\infty$$

(see e.g., [27], section 4.42), and the same argument holds in the case $\sum_{\rho_{\chi_1}: \gamma_{\chi_1}=0} \sum_{\rho_{\chi_2}}$, hence we can assume $\gamma_{\chi_j} \neq 0$, $j = 1, 2$. We start with the case $\gamma_{\chi_1} > 0$, $\gamma_{\chi_2} > 0$. By the Stirling's formula (3.5), we have

$$\begin{aligned} \sum_{\rho_{\chi_1} \in Z^+(\chi_1)} |f(\rho_{\chi_1})| \sum_{\rho_{\chi_2} \in Z^+(\chi_2)} |g(\rho_{\chi_2})| \left| \frac{\Gamma(\rho_{\chi_1}) \Gamma(\rho_{\chi_2})}{\Gamma(\rho_{\chi_1} + \rho_{\chi_2} + k + 1)} \right| \\ \ll \sum_{\rho_{\chi_1} \in Z^+(\chi_1)} |f(\rho_{\chi_1})| \sum_{\rho_{\chi_2} \in Z^+(\chi_2)} |g(\rho_{\chi_2})| \frac{1}{(\gamma_{\chi_1} + \gamma_{\chi_2})^{3/2+k}} \\ \ll \sum_{\rho_{\chi_1} \in Z^+(\chi_1)} \frac{|f(\rho_{\chi_1})|}{\gamma_{\chi_1}^{3/4+k/2}} \sum_{\rho_{\chi_2} \in Z^+(\chi_2)} \frac{|g(\rho_{\chi_2})|}{\gamma_{\chi_2}^{3/4+k/2}} \end{aligned}$$

by arithmetic-geometric mean, and both series converges absolutely if $k > 1/2$ since, by partial summation, taking $T > 0$, we have

$$(3.6) \quad \sum_{\rho_{\chi_1}: 0 < \gamma_{\chi_1} \leq T} \frac{|f(\rho_{\chi_1})|}{\gamma_{\chi_1}^{3/4+k/2}} \ll \sum_{\rho_{\chi_1}: 0 < \gamma_{\chi_1} \leq T} \left| \frac{f(\rho_{\chi_1})}{\rho_{\chi_1}} \right| \frac{1}{\gamma_{\chi_1}^{k/2-1/4}}$$

$$= \sum_{\rho_{\chi_1}: 0 < \gamma_{\chi_1} \leq T} \left| \frac{f(\rho_{\chi_1})}{\rho_{\chi_1}} \right| \frac{1}{T^{k/2-1/4}} + \frac{2k-1}{4} \int_{\delta_{\chi_1}}^T \sum_{\rho_{\chi_1}: 0 < \gamma_{\chi_1} \leq t} \left| \frac{f(\rho_{\chi_1})}{\rho_{\chi_1}} \right| t^{-k/2-3/4} dt$$

where $\delta_{\chi_1} > 0$ is a number such that

$$0 < \delta_{\chi_1} \leq \min \left\{ \gamma_{\chi_1} > 0 : L \left(\frac{1}{2} + i\gamma_{\chi_1}, \chi_1 \right) = 0 \right\}.$$

and the same considerations holds for the case $g(\rho_{\chi_2})$. Now, we consider the case $\gamma_{\chi_1} > 0$, $\gamma_{\chi_2} < 0$. Clearly, if $\rho_{\chi_2} = 1/2 + i\gamma_{\chi_2}$ is a zero of $L(s, \chi_2)$, then $\overline{\rho_{\chi_2}}$ is a zero of $L(s, \overline{\chi_2})$. Let us define

$$\rho_{\chi_2}^* := \overline{\rho_{\chi_2}}.$$

Then, we have to deal with

$$\sum_{\rho_{\chi_1} \in Z^+(\chi_1)} |f(\rho_{\chi_1})| \sum_{\rho_{\overline{\chi_2}} \in Z^+(\overline{\chi_2})} |g(\rho_{\chi_2}^*)| \left| \frac{\Gamma(\rho_{\chi_1}) \Gamma(\rho_{\chi_2}^*)}{\Gamma(\rho_{\chi_1} + \rho_{\chi_2}^* + k + 1)} \right|.$$

Let $0 < \alpha < 1$ be fixed. We split the second series in the following way:

$$\begin{aligned} \sum_{\rho_{\chi_2} \in Z^+(\overline{\chi_2})} &= \sum_{\substack{\rho_{\overline{\chi_2}} \in Z^+(\overline{\chi_2}) \\ |\gamma_{\chi_1} - \gamma_{\chi_2}| < \alpha \max(\gamma_{\chi_1}, \gamma_{\chi_2})}} + \sum_{\substack{\rho_{\overline{\chi_2}} \in Z^+(\overline{\chi_2}) \\ |\gamma_{\chi_1} - \gamma_{\chi_2}| \geq \alpha \max(\gamma_{\chi_1}, \gamma_{\chi_2})}} \\ &=: \sum_1 + \sum_2, \end{aligned}$$

say. We start with $\sum_1$. From the relation

$$|\Gamma(x + iy)| \geq \Gamma(x) \operatorname{sech}(\pi y)^{1/2}, \quad x \geq 1/2$$

(see [23], relation 5.6.7) we have

$$\begin{aligned} |\Gamma(\rho_{\chi_1} + \rho_{\chi_2}^* + k + 1)| &\geq \Gamma(2 + k) \min_{0 \leq t \leq \alpha \max(\gamma_{\chi_1}, \gamma_{\chi_2})} \operatorname{sech}(\pi t)^{1/2} \\ &\gg_k e^{-\alpha \pi (\gamma_{\chi_1} + \gamma_{\chi_2})/2} \end{aligned}$$

hence, using again (3.5), we get

$$\begin{aligned} &\sum_{\rho_{\chi_1} \in Z^+(\chi_1)} |f(\rho_{\chi_1})| \sum_{\substack{\rho_{\overline{\chi_2}} \in Z^+(\overline{\chi_2}) \\ |\gamma_{\chi_1} - \gamma_{\chi_2}| < \alpha \max(\gamma_{\chi_1}, \gamma_{\chi_2})}} |g(\rho_{\chi_2}^*)| \left| \frac{\Gamma(\rho_{\chi_1}) \Gamma(\rho_{\chi_2}^*)}{\Gamma(\rho_{\chi_1} + \rho_{\chi_2}^* + k + 1)} \right| \\ &\ll_k \sum_{\rho_{\chi_1} \in Z^+(\chi_1)} |f(\rho_{\chi_1})| e^{-\pi(1-\alpha)\gamma_{\chi_1}/2} \sum_{\rho_{\overline{\chi_2}} \in Z^+(\overline{\chi_2})} |g(\rho_{\chi_2}^*)| e^{-\pi(1-\alpha)\gamma_{\chi_2}/2} \end{aligned}$$

and the convergence follows again by partial summation. Now we consider $\sum_2$. We split in two further cases: if $|\gamma_{\chi_1} - \gamma_{\chi_2}| \geq \alpha \max(\gamma_{\chi_1}, \gamma_{\chi_2})$, then $\gamma_{\chi_1} \neq \gamma_{\chi_2}$, so if $\gamma_{\chi_1} > \gamma_{\chi_2}$, then again by (3.5)

$$\begin{aligned} & \sum_{\rho_{\chi_1} \in Z^+(\chi_1)} |f(\rho_{\chi_1})| \sum_{\substack{\rho_{\overline{\chi_2}} \in Z^+(\overline{\chi_2}) \\ |\gamma_{\chi_1} - \gamma_{\chi_2}| \geq \alpha \gamma_{\chi_1}}} |g(\rho_{\chi_2}^*)| \left| \frac{\Gamma(\rho_{\chi_1}) \Gamma(\rho_{\chi_2}^*)}{\Gamma(\rho_{\chi_1} + \rho_{\chi_2}^* + k + 1)} \right| \\ & \ll \sum_{\rho_{\chi_1} \in Z^+(\chi_1)} |f(\rho_{\chi_1})| \sum_{\substack{\rho_{\overline{\chi_2}} \in Z^+(\overline{\chi_2}) \\ |\gamma_{\chi_1} - \gamma_{\chi_2}| \geq \alpha \gamma_{\chi_1}}} |g(\rho_{\chi_2}^*)| \frac{e^{-\pi \gamma_{\chi_2}/2} e^{-\pi \gamma_{\chi_1}/2}}{(\gamma_{\chi_1} - \gamma_{\chi_2})^{3/2+k} e^{-\pi(\gamma_{\chi_1} - \gamma_{\chi_2})/2}} \\ & \ll_{\alpha} \sum_{\rho_{\chi_1} \in Z^+(\chi_1)} \frac{|f(\rho_{\chi_1})|}{\gamma_{\chi_1}^{3/2+k}} \sum_{\rho_{\overline{\chi_2}} \in Z^+(\overline{\chi_2})} |g(\rho_{\chi_2}^*)| e^{-\pi \gamma_{\chi_2}} \end{aligned}$$

and again the convergence follows. A similar calculation holds in the case $\gamma_{\chi_2} > \gamma_{\chi_1}$. $\square$

We also recall the following very easy fact, because we will use it very often.

Proposition 16. *For every $a > 0$, $\alpha \geq 0$ and $n \geq 0$ we have*

$$\int_0^a x^{\alpha} |\log(x)|^n dx \ll_n a^{\alpha+1} (1 + |\log(a)|^n).$$

3.4. Explicit formulae extended up to $x > 0$. In this part we derive explicit formulae for some classical arithmetic functions. For our purposes, it is important to formulate these identities for all $x > 0$, rather than only in the classical ranges $x > 1$ or $x > 2$. At first sight this may seem unnecessary, since for small values of x , the corresponding summatory functions are often trivial (for example, $\psi(x) := \sum_{n \leq x} \Lambda(n) = 0$ if $0 < x < 2$).

However, what is really needed is not so much an extension of the summatory functions themselves, but rather a formulation of the zero-sum side that remains meaningful when x is small, without having to rely on estimates for non-absolutely convergent series. For example, one has an explicit formula of the shape

$$(3.7) \quad \psi(x) = x - \sum_{\rho: |\gamma| \leq T} \frac{x^{\rho}}{\rho} + R(x, T)$$

valid for $x, T \geq 2$, for a suitable error $R(x, T)$. Even if the main term $x - \sum_{\rho: |\gamma| \leq T} \frac{x^{\rho}}{\rho}$ can formally be written also for $0 < x < 2$ such an extension is not harmless, since the sum over the zeros is not absolutely convergent and may produce an error term depending on T that worsens when T grows. Moreover, in our setting because, by Theorem 8 we are led to consider convolutions of the form

$$\int x^{\rho_1} (A - x)^{\rho_2} dx$$

and if the range of integration does not start at 0, then incomplete beta functions naturally appear, making the resulting expressions substantially less transparent than the corresponding full beta-type convolution.

We start by recalling the explicit formula for $\psi(x, \chi) := \sum_{n \leq x} \Lambda(n) \chi(n)$ and extending it to $x > 0$.

Theorem 17. (*Truncated explicit formula for $\psi(x, \chi)$*) Assume GRH. Let $x > 0$, $x \neq 1$, and $T \geq 2$. Let χ be a primitive non-principal character mod q , $q \in \mathbb{N}$, $q > 1$. Then

$$\psi(x, \chi) = - \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{x^{\rho_\chi}}{\rho_\chi} + R_\chi(x, T, q)$$

where ρ_χ runs over the non-trivial zeros of the function $L(s, \chi)$, with

$$R_\chi(x, T, q) \ll \log(q) + |\log(x)| + |\log(|x-1|)| + \frac{(x+1) \log(qT \max\{x, 1/x\})^2}{T}$$

and the implicit constant is absolute.

Proof. If $x > 1$ the claim follows from the classical truncated explicit formula. Indeed, defining

$$\psi_0(x, \chi) := \psi(x, \chi) - \frac{1}{2} \Lambda(x) \chi(x)$$

we have

$$\psi_0(x, \chi) = - \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{x^{\rho_\chi}}{\rho_\chi} - \frac{1}{2} \log(x-1) - \frac{\chi(-1)}{2} \log(x+1) + C(\chi) + R_\chi^1(x, T, q)$$

where

$$R_\chi^1(x, T, q) \ll \log(x) \min\left(1, \frac{x}{T \langle x \rangle}\right) + \frac{x}{T} \log(qTx)^2$$

and

$$C(\chi) := \frac{L'}{L}(1, \bar{\chi}) + \log\left(\frac{q}{2\pi}\right) - C_0$$

where C_0 is the Euler-Mascheroni constant (see [20], Corollary 12.11) and $\langle x \rangle$ means the distance of x to the nearest integer. The estimate $C(\chi) \ll \log(q)$ holds if there is no exceptional zero (see again Corollary 12.11). For the case $0 < x < 1$, we consider $y > 1$ and $\sigma_0 = 1/\log(y)$ we take

$$\begin{aligned} \sum'_{n \leq y} \frac{\bar{\chi}(n) \Lambda(n)}{n} &= -\frac{1}{2\pi i} \int_{\sigma_0 - iT}^{\sigma_0 + iT} \frac{L'}{L}(s+1, \bar{\chi}) \frac{y^s}{s} ds \\ &+ O\left(\frac{\log(y)}{y} \min\left(1, \frac{y}{T \langle y \rangle}\right) + \frac{\log(y)^2}{T}\right) \end{aligned}$$

by Theorem 5.2 and Corollary 5.3 of [20]. From now on, we denote with $\sigma := \operatorname{Re}(s)$, for $s \in \mathbb{C}$. Let κ be equal to 0 or 1 if $\bar{\chi}(-1) = 1$ or $\bar{\chi}(-1) = -1$, respectively, and let $\mathcal{A}(\kappa)$ be the set of points $s \in \mathbb{C}$ such that $\sigma \leq -1$ and $|s + 2n - \kappa| \geq 1/4$. Let $\mathcal{C}$ be the contour consisting of the line segments connecting $\sigma_0 - iT_1$, $-K - iT_1$, $-K + iT_1$ and $\sigma_0 + iT_1$ where K is chosen so that $K - \kappa$ is an odd positive integer and T_1 is chosen as in Lemma 12.9 of [20], in particular $T_1 \in [T, T+1]$. Now, for the horizontal segment, by Lemma 12.7 and 12.9 of [20], we have

$$\begin{aligned} &\ll \int_{-1+iT_1}^{\sigma_0+iT_1} \left| \frac{L'}{L}(s+1, \bar{\chi}) \frac{y^s}{s} \right| |ds| + \int_{-K+iT_1}^{-1+iT_1} \left| \frac{L'}{L}(s+1, \bar{\chi}) \frac{y^s}{s} \right| |ds| \\ &\ll \frac{\log(qT)^2}{T} \int_{-2}^{\sigma_0} y^\sigma d\sigma + \frac{\log(2qT)}{T} \int_{-\infty}^{-1} y^\sigma d\sigma \\ &\ll \frac{\log(qT)^2}{T \log(y)} + \frac{\log(2qT)}{T \log(y)} \end{aligned}$$

and since, from the vertical segment

$$\int_{-K-iT_1}^{-K+iT_1} \left| \frac{L'}{L}(s+1, \bar{\chi}) \frac{y^s}{s} \right| |ds| \ll y^{-K} \frac{\log(qK)T}{K} \rightarrow 0, \quad K \rightarrow +\infty$$

we get, by the residue theorem, that

$$\sum'_{n \leq y} \frac{\bar{\chi}(n) \Lambda(n)}{n} = - \sum_{\rho_{\bar{\chi}}: |\gamma_{\bar{\chi}}| < T} \frac{y^{\rho_{\bar{\chi}}-1}}{1 - \rho_{\bar{\chi}}} + \sum_{k \geq 1} \frac{y^{-2k-1+\kappa}}{2k+1-\kappa} - \frac{L'}{L}(1, \bar{\chi}) + \frac{\bar{\chi}(-1)+1}{2y} + R_{\bar{\chi}}^2(y, T, q)$$

with

$$R_{\bar{\chi}}^2(y, T, q) \ll \frac{\log(y)}{y} \min\left(1, \frac{y}{T \langle y \rangle}\right) + \frac{\log(qyT)^2}{T}.$$

Now, using the identities

$$\sum_{k \geq 1} \frac{y^{-2k}}{2k} = -\frac{1}{2} \log(1 - y^{-2}), \quad \sum_{k \geq 1} \frac{y^{-2k-1}}{2k+1} = \frac{1}{2} \log\left(\frac{y+1}{y-1}\right) - \frac{1}{y}$$

we are able to control the sum $\sum_{k \geq 1} \frac{y^{-2k-1+\kappa}}{2k+1-\kappa}$. Then, taking $y = 1/x$ and using the symmetries of the non-trivial zeros, that is, under GRH, if $\rho_{\bar{\chi}}$ is a non-trivial zero of $L(s, \bar{\chi})$, then $1 - \rho_{\bar{\chi}}$ is a zero of $L(s, \chi)$, we have that

$$\begin{aligned} - \sum_{\rho_{\chi}: |\gamma_{\chi}| < T} \frac{x^{\rho_{\chi}}}{\rho_{\chi}} &= O\left(\sum'_{n \leq 1/x} \frac{\Lambda(n)}{n} + \left| \frac{L'}{L}(1, \bar{\chi}) \right| + \left| \log\left(\frac{x-1}{x}\right) \right| \right. \\ &\quad \left. + |\log(x)| \left[1 + \min\left(1, \frac{x}{T \langle x \rangle}\right) + \min\left(1, \frac{1}{xT \langle 1/x \rangle}\right) \right] + \frac{\log(qT/x)^2}{T} \right) \end{aligned}$$

and, since $\left| \frac{L'}{L}(1, \chi) \right| = \left| \frac{L'}{L}(1, \bar{\chi}) \right| \ll \log(q)$ by Corollary 12.11 and observing that $\sum'_{n \leq 1/x} \frac{\Lambda(n)}{n} \ll |\log(x)|$ and $\psi_0(x, \chi) = 0$ if $0 < x < 1$, the thesis follows. $\square$

Due to the presence of the Möbius function and our use of explicit formulae, it is natural to introduce an additional conjectural assumption concerning the simplicity of the zeros of the Riemann zeta function. To address this issue, we recall the following conjecture.

Conjecture 18. (*Gonek-Hejhal conjecture* [11, 12]) *Let $\rho = \beta + i\gamma$ be a non-trivial zero of the Riemann zeta function $\zeta(s)$. Let*

$$J_k(T) := \sum_{0 < \gamma \leq T} \frac{1}{|\zeta'(\rho)|^{2k}}.$$

Then, for all $k \in \mathbb{R}$, we have

$$(3.8) \quad J_k(T) \asymp T \log(T)^{(k-1)^2}.$$

We recall that this conjecture implies that all non-trivial zeros of the Riemann zeta function are simple (see [15]). Note that some refinement by Hughes, Keating and O'Connell suggests that actually the conjecture is false if $k \geq 3/2$ (see again [14]). Actually, we just need the special case $k = 1$; for this reason, we use the following simpler conjecture.

Conjecture 19. *We have that*

$$J_1(T) = \sum_{0 < \gamma \leq T} \frac{1}{|\zeta'(\rho)|^2} \ll T.$$

Note that also this conjecture implies that all the zeros of the $\zeta(s)$ are simple (see [15] Conjecture 3.9). Moreover, we have, by the Cauchy-Schwarz inequality and the bound (3.3)

$$(3.9) \quad \sum_{0 < \gamma \leq T} \frac{1}{|\zeta'(\rho)|} \ll T \log(T)^{1/2}$$

which is off from (3.8) by an exponent $1/4$ on the $\log(T)$. It is not a real problem, for our aims. Hence, we can work directly with Conjecture 19.

Now we are able to recall the explicit formula for the Mertens' function $M(x) := \sum_{n \leq x} \mu(n)$. Also in this case, we need to extend the formula to all $x > 0$, excluding at most the case $x = 1$.

Theorem 20. (*Truncated explicit formula for $M(x)$*)

Let $0 < \varepsilon < 1/2$ be fixed. Assume RH, $x > 0$, $x \neq 1$, $U \geq 2$ and Conjecture 19. Then, we have

$$(3.10) \quad M(x) = \sum_{\rho: |\gamma| \leq U} \frac{x^\rho}{\zeta'(\rho)\rho} + O_\varepsilon \left(1 + x^\varepsilon + \frac{x(|\log(x)| + 1)}{U} + \frac{x - x^\varepsilon}{U^{1-\varepsilon} \log(x)} \right).$$

This formula can be found in [4] and, for $x \geq 2$, in [22], with some small differences in the error term, which however can be absorbed in (3.10). Note that, for our purposes, it is important to reduce the number of terms in the error term, in order to simplify the subsequent calculations. Indeed, we may actually write

$$(3.11) \quad M(x) = \sum_{\rho: |\gamma| \leq U} \frac{x^\rho}{\zeta'(\rho)\rho} + O \left(1 + x^\varepsilon + \frac{(x^\varepsilon + x)(|\log(x)| + 1)}{U^{1-\varepsilon}} \right)$$

where we use the elementary inequality

$$\frac{x - x^\varepsilon}{\log(x)} \leq x + x^\varepsilon, \quad x > 0, \quad x \neq 1.$$

Since we are interested to find an upper bound for weighted average of diagonal twisted EH problem also in the $\log$'s case, we also need an explicit formula for

$$\widetilde{M}(x) := \sum_{n \leq x} \mu(n) \log(n)$$

and, luckily, this can be derived by Theorem 20. Indeed, if $x > 1$, by partial summation,

$$(3.12) \quad \widetilde{M}(x) = \log(x) M(x) - \int_1^x \frac{M(t)}{t} dt$$

and inserting the explicit formula (3.11) we get

$$\widetilde{M}(x) = \sum_{\rho: |\gamma| \leq U} \frac{x^\rho}{\zeta'(\rho)\rho} \left(\log(x) - \frac{1}{\rho} \right) + O_\varepsilon \left(\log(x) + x^\varepsilon + \frac{(x^\varepsilon + x)(\log(x))}{U^{1-\varepsilon}} \right).$$

Now, if $0 < x < 1$, we trivially observe that

$$(3.13) \quad 0 = \widetilde{M}(x) = \log(x) M(x) = \log(x) M(x) - \sum_{\rho: |\gamma| \leq U} \frac{x^\rho}{\zeta'(\rho)\rho^2} + O_\varepsilon \left(1 + |\log(x)| + x^\varepsilon + \frac{(x^\varepsilon + x)(|\log(x)| + 1)}{U^{1-\varepsilon}} \right)$$

since $\sum_{\rho} \frac{1}{|\zeta'(\rho)\rho^2|}$ is absolutely convergent, by Conjecture 19, and so its contribute is absorbed in the $O_{\varepsilon}(1)$ term. Then, inserting the explicit formula in (3.13) and recalling that, if $0 < x < 1$, $x^A |\log(x)|^B \ll 1$ for every fixed $A, B > 0$, we get

$$(3.14) \quad \widetilde{M}(x) = \sum_{\rho: |\gamma| \leq U} \frac{x^{\rho}}{\zeta'(\rho)\rho} \left(\log(x) - \frac{1}{\rho} \right) + O_{\varepsilon} \left(1 + |\log(x)| + x^{\varepsilon} + \frac{(x^{\varepsilon} + x)(|\log(x)| + 1)}{U^{1-\varepsilon}} \right)$$

valid for $x > 0$, $x \neq 1$.

4. WEIGHTED AVERAGES OF DIAGONAL TWISTED EH PROBLEM WITH $g(n) = \mu(n)$

Now we are able to study general weighted sum

$$\sum_n \sum_m \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right)$$

where $N \geq 4$ is a natural number. We divide this section in two parts: in the first one, we will use weights in the Sobolev space $W^{2,1}$, and this regularity allows to get a bound for the diagonal twisted EH for all $0 < \theta < 1$; in the second part, we will use weights that belong to the Hölder-Zygmud spaces $\mathcal{C}^{\delta}$, $\delta \in [1, 2]$, and we pay the minor regularity with a smaller range of θ .

4.1. Weight in Sobolev space $W^{2,1}$. Let $f : \mathbb{R} \rightarrow \mathbb{C}$ be a function that verifies the hypotheses of Theorem 8. Since we are interested in weights that have support in $\mathbb{R}_0^+$, From now on we will assume that $\alpha = 0$ and $\beta \in \mathbb{R}^+$. This choice also has the advantage of being able to use the Theorem 8 and Theorem 9 without boundary terms and without creating dependencies between α, β and N . The price to pay will be having to assume that the second derivative of f is bounded in a neighborhood of the origin, due to the logarithmic terms that will emerge from the various estimates.

We recall that hypotheses of Theorem 8 imply, essentially, that $f|_{(0,\beta)} \in W^{2,1}(0, \beta)$. We have, taking $g_1(n) = \Lambda(n)\chi(n)$, $g_2(m) = \mu(m)$, that

$$\sum_{n \leq N\beta} \sum_{m \leq N\beta-n} \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right) = \frac{1}{N} \int_0^{\beta} f''(w) \int_0^{Nw} M(s) \psi(Nw-s, \chi) ds dw$$

so we have

$$\begin{aligned} & \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta-n} \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right) \\ &= \frac{1}{N\varphi(q)} \int_0^{\beta} f''(w) \int_0^{Nw} M(s) \sum_{\chi \neq \chi_0} \bar{\chi}(N) \psi(Nw-s, \chi) ds dw. \end{aligned}$$

We recall the abuse of notation: the sum $\sum_{\chi \neq \chi_0}$ is intended on all characters $\pmod{q}$ other than the principal one χ_0 but $\chi(n)$ is the primitive character that induces χ . So, if we apply the previous relation to the weighted average of the diagonal twisted EH problem, we get

$$\sum_{\substack{1 < q \leq N^{\theta} \\ (N,q)=1}} \frac{1}{\varphi(q)} \left| \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta-n} \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right) \right|$$

$$(4.1) \quad = \sum_{\substack{1 < q \leq N^\theta \\ (N, q) = 1}} \frac{1}{\varphi(q)} \left| \frac{1}{N} \int_0^\beta f''(w) \int_0^{Nw} M(s) \sum_{\chi \neq \chi_0} \bar{\chi}(N) \psi(Nw - s, \chi) ds dw \right|.$$

For studying this object, we firstly prove an explicit formula for the weighted average.

Before going into details, let's point out that in the following proof, given the complexity of the calculations, we will prioritize clarity over the actual logarithmic contribution of the error term: this is because, for our purposes, it will not significantly influence the final result.

Theorem 21. Fix $0 < \varepsilon < 1/2$ and assume GRH and Conjecture 19. Let $N \geq 4$, $q \in \mathbb{N}$, $1 < q < N^\theta$, $0 < \theta < 1$. Moreover, let $\beta > 0$ and $f : \mathbb{R} \rightarrow \mathbb{C}$ satisfying the hypotheses of Theorem 8 with f'' bounded in a neighborhood of the origin. Then

$$\begin{aligned} & \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta - n} \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right) \\ &= -\frac{1}{\varphi(q)} \sum_{\rho: |\gamma| \leq U} \frac{N^\rho \Gamma(\rho)}{\zeta'(\rho)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{N^{\rho_\chi} \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \int_0^\beta f''(w) w^{\rho + \rho_\chi + 1} dw \\ &+ O_\varepsilon \left(\frac{1}{N} \int_0^\beta |f''(w)| [\mathcal{E}_1(N, w, q, U, T) + \mathcal{E}_2(N, w, q, U, T)] dw \right) \end{aligned}$$

where

$$\begin{aligned} \mathcal{E}_1(N, w, q, U, T) &:= \varphi(q) N^{3/2+\varepsilon} \log(q)^3 \log(N)^3 \left(1 + w^{3/2+\varepsilon}\right) \left(1 + |\log(w)|^3\right) \\ &\times \left[1 + \frac{N(1+w) \log(qT)^2}{T} + \frac{N^{1-\varepsilon} (w^{1-\varepsilon} + 1)}{U^{1-\varepsilon}} \right], \\ \mathcal{E}_2(N, w, q, U, T) &= \varphi(q) N^{1+\varepsilon} \log(q) \log(N)^3 \left(1 + |\log(w)|^3\right) \\ &\times (1 + w^{1+\varepsilon}) \left[1 + \frac{N \log(qT)^2}{T} \right] \left[1 + \frac{N^{1-1\varepsilon}}{U^{1-\varepsilon}} \right]. \end{aligned}$$

and the implicit constant depends only on ε .

Proof. By (4.1), the problem boils down to considering

$$\frac{1}{N} \int_0^\beta f''(w) \int_0^{Nw} M(s) \sum_{\chi \neq \chi_0} \bar{\chi}(N) \psi(Nw - s, \chi) ds dw.$$

From the explicit formulas we write $\psi(x, \chi)$ and $M(x)$ as

$$\psi(x, \chi) = - \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{x^{\rho_\chi}}{\rho_\chi} + R_{1,\chi}(x, T, q), \quad M(x) = \sum_{\rho: |\gamma| \leq U} \frac{x^\rho}{\zeta'(\rho) \rho} + R_2(x, U)$$

then, putting

$$\mathcal{M}_{1,\chi}(x, T) := - \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{x^{\rho_\chi}}{\rho_\chi}, \quad \mathcal{M}_2(x, U) := \sum_{\rho: |\gamma| \leq U} \frac{x^\rho}{\zeta'(\rho) \rho}$$

and observing

$$\mathcal{M}_{1,\chi}(x, T) = \psi(x, \chi) - R_{1,\chi}(x, T, q), \quad \mathcal{M}_2(x, U) = M(x) - R_2(x, U)$$

then we have the decomposition

$$(4.2) \quad \int_0^{Nw} M(s) \sum_{\chi \neq \chi_0} \bar{\chi}(N) \psi(Nw - s, \chi) ds = \int_0^{Nw} \mathcal{M}_2(s, U) \sum_{\chi \neq \chi_0} \bar{\chi}(N) \mathcal{M}_{1,\chi}(Nw - s, T) ds$$

$$(4.3) \quad + \int_0^{Nw} M(s) \sum_{\chi \neq \chi_0} \bar{\chi}(N) R_{1,\chi}(Nw - s, T, q) ds + \int_0^{Nw} R_2(s, U) \sum_{\chi \neq \chi_0} \bar{\chi}(N) \psi(Nw - s, \chi) ds$$

$$(4.4) \quad - \int_0^{Nw} R_2(s, U) \sum_{\chi \neq \chi_0} \bar{\chi}(N) R_{1,\chi}(Nw - s, T, q) ds.$$

It is clear that, from the main terms, we get

$$\begin{aligned} & \frac{1}{N} \int_0^\beta f''(w) \int_0^{Nw} \mathcal{M}_2(s, U) \mathcal{M}_{1,\chi}(Nw - s, T) ds dw \\ &= -\frac{1}{N} \sum_{\rho: |\gamma| \leq U} \frac{1}{\zeta'(\rho)\rho} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{1}{\rho_\chi} \int_0^\beta f''(w) \int_0^{Nw} s^\rho (Nw - s)^{\rho_\chi} ds dw \\ &= - \sum_{\rho: |\gamma| \leq U} \frac{N^\rho \Gamma(\rho)}{\zeta'(\rho)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{N^{\rho_\chi} \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \int_0^\beta f''(w) w^{\rho + \rho_\chi + 1} dw \end{aligned}$$

so we now focus on the error terms.

Under GRH, we know that

$$M(x) \ll_\varepsilon x^{1/2+\varepsilon}, \quad \psi(x, \chi) \ll x^{1/2} \log(x) \log(qx)$$

for $x \geq 2$ and $q \geq 1$ (see [28] p. 371 and Theorem 13.7 of [20]), hence if $x \leq CN$ for some fixed $C > 0$, we have the uniform bounds, actually valid for every $0 < x \leq CN$

$$(4.5) \quad M(x) \ll N^{1/2+\varepsilon}, \quad \psi(x, \chi) \ll N^{1/2} \log(N) \log(qN).$$

Let us define

$$\begin{aligned} I_1 &= I_1(w, N, q, T) := \int_0^{Nw} M(s) \sum_{\chi \neq \chi_0} \bar{\chi}(N) R_{1,\chi}(Nw - s, T, q) ds \\ I_2 &= I_2(w, N, q, U) := \int_0^{Nw} R_2(s, U) \sum_{\chi \neq \chi_0} \bar{\chi}(N) \psi(Nw - s, \chi) ds \\ I_3 &= I_3(w, N, q, T, U) := \int_0^{Nw} R_2(s, U) \sum_{\chi \neq \chi_0} \bar{\chi}(N) R_{1,\chi}(Nw - s, T, q) ds. \end{aligned}$$

Bounding I_1 .

By Theorem 17, since $\log(q) \geq \log(2) \gg 1$ and since $N \geq 4$, we have

$$\begin{aligned} I_1 &\ll_\varepsilon \varphi(q) N^{1/2+\varepsilon} w^{1/2+\varepsilon} \int_0^{Nw} [\log(q) + |\log(s)| + |\log(|s-1|)|] ds \\ &\quad + \frac{\varphi(q) N^{1/2+\varepsilon} w^{1/2+\varepsilon}}{T} \int_0^{Nw} (s+1) \left[\log(qsT)^2 + \log(qT/s)^2 \right] ds \end{aligned}$$

and using Proposition 16 several times, we get

$$I_1 \ll_\varepsilon \varphi(q) \log(q) N^{3/2+\varepsilon} w^{3/2+\varepsilon} (1 + |\log(Nw)|)$$

$$\begin{aligned}
& + \frac{\varphi(q) N^{3/2+\varepsilon} w^{3/2+\varepsilon} \log(qT)^2 (1+Nw) \left(1 + \log(Nw)^2\right)}{T}. \\
& \ll_{\varepsilon} \varphi(q) \log(q) N^{3/2+\varepsilon} w^{3/2+\varepsilon} \left[1 + |\log(Nw)| + \frac{(1+Nw) \log(qT)^2 \left(1 + \log(Nw)^2\right)}{T} \right] \\
& \ll_{\varepsilon} \varphi(q) \log(q) N^{3/2+\varepsilon} \log(N)^2 w^{3/2+\varepsilon} \left(1 + \log(w)^2\right) \left[1 + \frac{N(1+w) \log(qT)^2}{T} \right]
\end{aligned}$$

Bounding I_2 .

By Theorem 20 and recalling again the elementary bound

$$\frac{s - s^{\varepsilon}}{\log(s)} \leq s + s^{\varepsilon}, \quad s > 0, s \neq 1$$

we have

$$(4.6) \quad I_2 \ll_{\varepsilon} \varphi(q) N^{1/2} w^{1/2} |\log(Nw)| |\log(qNw)| \int_0^{Nw} [1 + s^{\varepsilon}] ds$$

$$(4.7) \quad + \varphi(q) N^{1/2} w^{1/2} |\log(wN)| |\log(qNw)| \int_0^{Nw} \left[\frac{(s + s^{\varepsilon})(|\log(s)| + 1)}{U^{1-\varepsilon}} \right] ds$$

and again from Proposition 16 we get

$$\begin{aligned}
& I_2 \ll_{\varepsilon} \varphi(q) N^{3/2+\varepsilon} w^{3/2} (1 + w^{\varepsilon}) |\log(Nw)| |\log(qNw)| \\
& + \frac{\varphi(q) N^{5/2} |\log(Nw)| |\log(qNw)| (|\log(Nw)| + 1) w^{1/2+\varepsilon} (w^{2-\varepsilon} + 1)}{U^{1-\varepsilon}} \\
& \ll_{\varepsilon} \varphi(q) N^{3/2+\varepsilon} \left(\log(q)^2 + \log(N)^2 \right) \left(1 + \log(w)^2 \right) w^{3/2} (1 + w^{\varepsilon}) \\
& + \frac{\varphi(q) N^{5/2} \left(\log(q)^3 + \log(N)^3 \right) \left(1 + |\log(w)|^3 \right) w^{1/2+\varepsilon} (w^{2-\varepsilon} + 1)}{U^{1-\varepsilon}}
\end{aligned}$$

and since

$$\log(w)^2 \ll 1 + |\log(w)|^3, \quad \forall w > 0$$

and

$$\log(q)^3 + \log(N)^3 \ll \log(q)^3 \log(N)^3$$

we get

$$\begin{aligned}
& I_2 \ll_{\varepsilon} \varphi(q) N^{3/2+\varepsilon} \log(q)^3 \log(N)^3 \\
& \times \left[w^{3/2} (1 + w^{\varepsilon}) \left(1 + \log(w)^2 \right) + \frac{N^{1-\varepsilon} \left(1 + |\log(w)|^3 \right) w^{1/2+\varepsilon} (w^{2-\varepsilon} + 1)}{U^{1-\varepsilon}} \right]
\end{aligned}$$

Bounding I_3 .

We have, by the Cauchy-Schwarz inequality, that

$$I_3 \ll_{\varepsilon} \sum_{\chi \neq \chi_0} \left(\int_0^{Nw} |R_2(s, U)|^2 ds \right)^{1/2} \left(\int_0^{Nw} |R_{1,\chi}(s, T, q)|^2 ds \right)^{1/2}$$

and using the trivial bound $(a_1 + \dots + a_n)^2 \ll_n a_1^2 + \dots + a_n^2$ and Proposition 16, we conclude

$$\int_0^{Nw} |R_{1,\chi}(s, T, q)|^2 ds \ll \int_0^{Nw} \left[\log(q)^2 + \log(s)^2 + |\log(|s-1|)|^2 \right] ds$$

$$\begin{aligned}
& + \frac{1}{T^2} \int_0^{Nw} (s+1)^2 \left[\log(qsT)^4 + \log(qT/s)^4 \right] ds \\
& \ll N(1+w) \log(q)^2 \log(N)^4 \left[\left(1 + \log(w)^2\right) + \frac{N^2(1+w^2) \log(qT)^4 (1 + \log(w)^4)}{T^2} \right]
\end{aligned}$$

and for $R_2(s, U)$, using the same tools of the previous bound, we obtain

$$\begin{aligned}
\int_0^{Nw} R_2(s, U)^2 ds & \ll_\varepsilon \int_0^{Nw} \left[1 + s^{2\varepsilon} + \left[\frac{(s^2 + s^{2\varepsilon}) (\log(s)^2 + 1)}{U^{2-2\varepsilon}} \right] \right] ds \\
& \ll_\varepsilon N^{1+2\varepsilon} (1 + w^{1+2\varepsilon}) + \frac{N^3 (1 + w^3) (1 + \log(Nw)^2)}{U^{2-2\varepsilon}} \\
& \ll_\varepsilon N^{1+2\varepsilon} \log(N)^2 (1 + w^{1+2\varepsilon}) \left[1 + \frac{N^{2-2\varepsilon} (1 + w^{2-2\varepsilon}) (1 + \log(w)^2)}{U^{2-2\varepsilon}} \right]
\end{aligned}$$

and finally, exploiting the subadditivity of the square root, we get

$$\begin{aligned}
I_3 & \ll_\varepsilon \varphi(q) N^{1+\varepsilon} (1 + w^{1+\varepsilon}) \log(q) \log(N)^3 (1 + |\log(w)|^3) \\
& \quad \times \left[1 + \frac{N(1+w) \log(qT)^2}{T} \right] \left[1 + \frac{N^{1-\varepsilon} (1 + w^{1-\varepsilon})}{U^{1-\varepsilon}} \right] \\
& \ll_\varepsilon \varphi(q) N^{1+\varepsilon} \log(q) \log(N)^3 (1 + |\log(w)|^3) (1 + w^{1+\varepsilon}) \left[1 + \frac{N \log(qT)^2}{T} \right] \left[1 + \frac{N^{1-\varepsilon}}{U^{1-\varepsilon}} \right]
\end{aligned}$$

Now, in view of reducing the complexity of the calculations, it is useful to get a common upper bound for I_1 and I_2 . Observe that, since

$$(1 + |\log(A)|^k) \ll (1 + |\log(A)|^n)$$

if $n \geq k$, for every $A > 0$, then, combining the pieces, we get

$$\begin{aligned}
I_1, I_2 & \ll_\varepsilon \varphi(q) N^{3/2+\varepsilon} \log(q)^3 \log(N)^3 (1 + w^{3/2+\varepsilon}) (1 + |\log(w)|^3) \\
& \quad \times \left[1 + \frac{N(1+w) \log(qT)^2}{T} + \frac{N^{1-\varepsilon} (w^{1-\varepsilon} + 1)}{U^{1-\varepsilon}} \right].
\end{aligned}$$

This concludes the proof $\square$

From the previous explicit formula now we finally obtain a bound for the weighted average of diagonal twisted EH problem.

Theorem 22. *Fix $0 < \varepsilon < 1/2$ and assume all the hypotheses of the previous theorem with $\theta = 1 - 2\varepsilon$. Then, the following weighted average version of the diagonal twisted Elliott–Halberstam conjecture holds*

$$\begin{aligned}
& \sum_{\substack{1 < q \leq N^{1-2\varepsilon} \\ (N, q) = 1}} \frac{1}{\varphi(q)} \left| \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta - n} \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right) \right| \\
& \ll_\varepsilon N^{2-\varepsilon} E(f'')
\end{aligned}$$

where

$$E(f'') := \int_0^\beta |f''(w)| \left(1 + |\log(w)|^3\right) (1 + w^2) dw.$$

and the implicit constant depends only on ε .

Proof. From Theorem 21 we know that the main average admits an explicit formula, so the problem boils down to evaluate every single term of it. We begin by estimating the main term. We have, fixing $\varepsilon > 0$, that

$$(4.8) \quad \left| \sum_{\rho: |\gamma| \leq U} \frac{N^\rho \Gamma(\rho)}{\zeta'(\rho)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{N^{\rho_\chi} \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \int_0^\beta f''(w) w^{\rho + \rho_\chi + 1} dw \right| \\ \leq N \varphi(q) \int_0^\beta |f''(w)| w^2 dw \max_{\chi \bmod q} \sum_{\rho: |\gamma| \leq U} \frac{1}{|\zeta'(\rho)|} \sum_{\rho_\chi: |\gamma_\chi| \leq T} \left| \frac{\Gamma(\rho) \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \right|.$$

Now, by Theorem 15 we observe that the double series converges absolutely, since $f(z) := \frac{1}{|\zeta'(z)|}$, $g(z) \equiv 1$ verify the hypotheses, by Conjecture 19, so we may let $T, U \rightarrow +\infty$. As for the dependence on q , we observe, starting considering $\gamma_\chi > 4$, that

$$\sum_{\rho} \frac{1}{|\zeta'(\rho)|} \sum_{\rho_\chi: \gamma_\chi > 4} \left| \frac{\Gamma(\rho) \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \right| = \sum_{\rho} \frac{1}{|\zeta'(\rho)|} \sum_{k \geq 2} \sum_{\rho_\chi: 2^k < \gamma_\chi \leq 2^{k+1}} \left| \frac{\Gamma(\rho) \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \right|$$

and since, if $T \geq 4$ and $q > 1$, we have, by (3.4), that

$$\sum_{\rho_\chi: 2^k < \gamma_\chi \leq 2^{k+1}} \left| \frac{\Gamma(\rho) \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \right| \ll \left(N(2^{k+1}, \chi) - N(2^k, \chi) \right) \max_{\rho_\chi: 2^k < \gamma_\chi \leq 2^{k+1}} \left| \frac{\Gamma(\frac{1}{2} + i\gamma) \Gamma(\frac{1}{2} + i\gamma_\chi)}{\Gamma(3 + i(\gamma + \gamma_\chi))} \right|.$$

Now, by Proposition 13 we deduce that

$$\left| \frac{\Gamma(\frac{1}{2} + i\gamma) \Gamma(\frac{1}{2} + i\gamma_\chi)}{\Gamma(3 + i(\gamma + \gamma_\chi))} \right| \ll \begin{cases} \frac{e^{\pi(|\gamma + \gamma_\chi| - |\gamma| - \gamma_\chi)/2}}{|\gamma + \gamma_\chi|^{5/2}}, & |\gamma + \gamma_\chi| > 1 \\ e^{-\pi(|\gamma| + \gamma_\chi)/2}, & |\gamma + \gamma_\chi| \leq 1 \end{cases}$$

hence it is enough to consider the “worst” case $\gamma, \gamma_\chi > 0$. In such case, by (3.4), we get

$$\sum_{\rho: \gamma > 0} \frac{1}{|\zeta'(\rho)|} \sum_{k \geq 2} \sum_{\rho_\chi: 2^k < \gamma_\chi \leq 2^{k+1}} \left| \frac{\Gamma(\rho) \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \right| \ll \log(q) \sum_{\rho: \gamma > 0} \frac{1}{|\zeta'(\rho)|} \sum_{k \geq 2} \frac{2^k \log(2^k)}{(\gamma + 2^k)^{5/2}} \\ \ll \log(q) \sum_{\rho: \gamma > 0} \frac{1}{|\zeta'(\rho)| \gamma^{5/4}} \sum_{k \geq 2} \log(k) 2^{-k/4} \ll \log(q)$$

the remaining cases, that is $|\gamma_\chi| \leq 4$ or $\gamma_\chi < -4$ or $\gamma < 0$ can be treated similarly. Hence, the main term can be bounded by

$$\ll N \varphi(q) \log(q) \int_0^\beta |f''(w)| w^2 dw$$

where the implicit constant is absolute. It remain to observe that, if $T, U \rightarrow +\infty$, the error terms are

$$\lim_{T, U \rightarrow +\infty} \mathcal{E}_1(N, w, q, U, T) = \varphi(q) N^{3/2+\varepsilon} \log(q)^3 \log(N)^3 \left(1 + w^{3/2+\varepsilon}\right) \left(1 + |\log(w)|^3\right)$$

$$\lim_{T, U \rightarrow +\infty} \mathcal{E}_2(N, w, q, U, T) = \varphi(q) N^{1+\varepsilon} \log(q) \log(N)^3 \left(1 + |\log(w)|^3\right) (1 + w^2)$$

Therefore, these terms produce the error

$$E_{\max} := \varphi(q) N^{1/2+\varepsilon} \log(q)^3 \log(N)^3 \int_0^\beta |f''(w)| \left(1 + |\log(w)|^3\right) (1+w^2) dw$$

hence we get, taking $\theta = 1 - 2\varepsilon$,

$$\begin{aligned} & \sum_{\substack{1 < q \leq N^{1-2\varepsilon} \\ (N,q)=1}} \frac{1}{\varphi(q)} \left| \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta-n} \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right) \right| \\ & \ll_\varepsilon N \int_0^\beta |f''(w)| w^2 dw \sum_{q \leq N^{1-2\varepsilon}} \log(q) \\ & + N^{1/2+\varepsilon} \int_0^\beta |f''(w)| \left(1 + |\log(w)|^3\right) (1+w^2) dw \sum_{q \leq N^{1-2\varepsilon}} \log(q)^3 \\ & \ll_\varepsilon \left(N^{2-2\varepsilon} \log(N) + \log(N)^3 N^{3/2} \right) E(f'') \ll_\varepsilon N^{2-\varepsilon} E(f'') \end{aligned}$$

where

$$E(f'') := \int_0^\beta |f''(w)| \left(1 + |\log(w)|^3\right) (1+w^2) dw.$$

So we finally get

$$\begin{aligned} & \sum_{\substack{1 < q \leq N^{1-2\varepsilon} \\ (N,q)=1}} \frac{1}{\varphi(q)} \left| \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta-n} \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right) \right| \\ & \ll_\varepsilon N^{2-\varepsilon} E(f''). \end{aligned}$$

□

4.2. Weights in Hölder-Zygmund class. As we said in the previous sections, the hypothesis $f|_{(0,\beta)} \in W^{2,1}(0,\beta)$ is, morally, minimal if we want to use Theorem 8, but this is no longer the case for the discrete version of that theorem. In the latter case, we are able to work with more general functions, and we will show that the regularity offered by the Hölder-Zygmund class is sufficient. Again, we first need an explicit formula for the weighted average.

Theorem 23. Fix $0 < \varepsilon < 1/2$. Assume GRH and the Conjecture 19. Let $N \geq 4$, $T, U \geq 4$, $q \in \mathbb{N}$, $1 < q < N^\theta$, $0 < \theta < 1$. Moreover, let $\beta > 0$ and $f : \mathbb{R} \rightarrow \mathbb{C}$ satisfy the hypotheses of Theorem 9. Fix $\delta \in [1, 2)$ and assume $f \in \mathcal{C}^\delta(\mathbb{R})$. Then

$$\begin{aligned} & \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta-n} \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right) \\ & = -\frac{1}{\varphi(q)} \sum_{\rho: |\gamma| \leq U} \frac{\Gamma(\rho)}{\zeta'(\rho)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{\Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \\ & \quad \times \sum_{k \leq N\beta-1} \Delta_{1/N}^2 \left(f, \frac{k}{N} \right) k^{\rho+\rho_\chi+1} \\ & \quad + O_{f,\beta,\varepsilon,\theta} \left(N^{5/2-\delta} \log(qT)^2 + \frac{N^{3-\delta} \log(qNT)^2}{T} \right) \end{aligned}$$

$$+ \log(q)^3 N^{5/2-\delta+\varepsilon} \log(N)^3 \left[1 + \frac{N \log(qT)^2}{T} \right] \left[1 + \frac{N^{1-\varepsilon}}{U^{1-\varepsilon}} \right]$$

Proof. We fix $\delta \in [1, 2)$ and $f \in \mathcal{C}^\delta(\mathbb{R})$, then we have

$$\Delta_{1/N}^2 \left(f, \frac{k}{N} \right) \ll_f N^{-\delta}$$

by Theorem 12. From this fact and from Corollary 10, we have

$$\begin{aligned} & \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta-n} \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right) \\ &= \frac{1}{\varphi(q)} \sum_{k \leq N\beta-1} \Delta_{1/N}^2 \left(f, \frac{k}{N} \right) \sum_{\chi \neq \chi_0} \bar{\chi}(N) \int_0^k \psi(s, \chi) M(k-s) ds \\ & \quad + O_{f,\beta} \left(\frac{N^{-\delta}}{\varphi(q)} \sum_{k \leq N\beta-1} \sum_{\chi \neq \chi_0} \left| \sum_{u=0}^k \psi(u, \chi) \mu(k-u) \right| \right). \end{aligned}$$

Using the explicit formulas and the calculations of Theorem 21 and the same notation, we get

$$\begin{aligned} & \int_0^k M(s) \sum_{\chi \neq \chi_0} \bar{\chi}(N) \psi(k-s, \chi) ds = - \sum_{\rho: |\gamma| \leq U} \frac{\Gamma(\rho)}{\zeta'(\rho)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{\Gamma(\rho_\chi) k^{\rho+\rho_\chi+1}}{\Gamma(\rho_\chi + \rho + 2)} \\ & \quad + O_\varepsilon(I_1(k/N, N, q, T) + I_2(k/N, N, q, U) + I_3(k/N, N, q, T, U)). \end{aligned}$$

So we obtain

$$\begin{aligned} & \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta-n} \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right) \\ &= -\frac{1}{\varphi(q)} \sum_{\rho: |\gamma| \leq U} \frac{\Gamma(\rho)}{\zeta'(\rho)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{\Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \sum_{k \leq N\beta-1} \Delta_{1/N}^2 \left(f, \frac{k}{N} \right) k^{\rho+\rho_\chi+1} \\ & \quad + O_{f,\varepsilon} \left(\frac{N^{-\delta}}{\varphi(q)} \sum_{k \leq N\beta-1} [I_1(k/N, N, q, T) + I_2(k/N, N, q, U) + I_3(k/N, N, q, T, U)] \right) \\ & \quad + O_{f,\beta} \left(\frac{N^{-\delta}}{\varphi(q)} \sum_{k \leq N\beta-1} \sum_{\chi \neq \chi_0} \left| \sum_{u=0}^k \psi(u, \chi) \mu(k-u) \right| \right). \end{aligned}$$

We begin by estimating I_1, I_2 . Since we proved in Theorem 21 that

$$\begin{aligned} I_1, I_2 &\ll \varphi(q) N^{3/2+\varepsilon} \log(q)^3 \log(N)^3 \left(1 + w^{3/2+\varepsilon} \right) \left(1 + |\log(w)|^3 \right) \\ &\quad \times \left[1 + \frac{N(1+w) \log(qT)^2}{T} + \frac{N^{1-\varepsilon} (w^{1-\varepsilon} + 1)}{U^{1-\varepsilon}} \right] \end{aligned}$$

but, in this case, the convolution is from 0 to k instead of 0 to Nw , hence we have to consider

$$\begin{aligned} & \log(q)^3 \varphi(q) N^{-\delta} \sum_{k \leq N\beta-1} k^{3/2+\varepsilon} \log(k)^3 \left[1 + \frac{k \log(qT)^2}{T} + \frac{k^{1-\varepsilon}}{U^{1-\varepsilon}} \right] \\ & \ll_\beta \log(q)^3 \varphi(q) N^{5/2-\delta+\varepsilon} \log(N)^3 \left[1 + \frac{N \log(qT)^2}{T} + \frac{N^{1-\varepsilon}}{U^{1-\varepsilon}} \right]. \end{aligned}$$

In the same spirit, taking I_3 , we have to consider

$$\begin{aligned} & \log(q) \varphi(q) N^{-\delta} \sum_{k \leq N\beta-1} k^{1+\varepsilon} \log(k)^3 \left[1 + \frac{k \log(qT)^2}{T} \right] \left[1 + \frac{k^{1-\varepsilon}}{U^{1-\varepsilon}} \right] \\ & \ll_{\beta} \varphi(q) \log(q) N^{2+\varepsilon-\delta} \log(N)^3 \left[1 + \frac{N \log(qT)^2}{T} \right] \left[1 + \frac{N^{1-\varepsilon}}{U^{1-\varepsilon}} \right]. \end{aligned}$$

Now, we consider the second error term. We have

$$\frac{N^{-\delta}}{\varphi(q)} \sum_{k \leq N\beta-1} \sum_{\chi \neq \chi_0} \left| \sum_{u=0}^k \psi(u, \chi) \mu(k-u) \right|.$$

Inserting the explicit formula of $\psi(u, \chi)$ proved in Theorem 17, we get, recalling that $q < N$,

$$\begin{aligned} & \frac{N^{-\delta}}{\varphi(q)} \sum_{k \leq N\beta-1} \sum_{\chi \neq \chi_0} \left| \sum_{\rho_{\chi}: |\gamma_{\chi}| \leq T} \frac{1}{\rho_{\chi}} \sum_{u=1}^{k-1} u^{\rho_{\chi}} \mu(k-u) \right| \\ & + O_{\beta, \theta} \left(N^{2-\delta} \log(N) + N^{-\delta} \sum_{k \leq N\beta-1} \sum_{u=1}^{k-1} \frac{u \log(quT)^2}{T} \right). \end{aligned}$$

Clearly

$$N^{-\delta} \sum_{k \leq N\beta-1} \sum_{u=1}^{k-1} \frac{u \log(quT)^2}{T} \ll \frac{N^{3-\delta} \log(qNT)^2}{T}$$

and from the main term we get

$$\begin{aligned} & \frac{N^{-\delta}}{\varphi(q)} \sum_{k \leq N\beta-1} \sum_{\chi \neq \chi_0} \left| \sum_{\rho_{\chi}: |\gamma_{\chi}| \leq T} \frac{1}{\rho_{\chi}} \sum_{u=1}^{k-1} u^{\rho_{\chi}} \mu(k-u) \right| \\ & \ll \frac{N^{-\delta}}{\varphi(q)} \sum_{k \leq N\beta-1} k^{3/2} \sum_{\chi \neq \chi_0} \sum_{\rho_{\chi}: |\gamma_{\chi}| \leq T} \frac{1}{|\rho_{\chi}|} \ll N^{5/2-\delta} \log(qT)^2 \end{aligned}$$

from the classical results

$$\sum_{\rho_{\chi}: |\gamma_{\chi}| \leq T} \frac{1}{|\rho_{\chi}|} \ll \log(qT)^2$$

which can be proved using (3.4). The claim follows by combining the previous estimates. $\square$

Now we are able to prove the main theorem of this subsection.

Theorem 24. *Under the hypotheses of the previous theorem and taking $\theta = \delta - 1 - 2\varepsilon$ with a fixed $0 < \varepsilon < (\delta - 1)/2$ if $\delta \in [\frac{3}{2}, 2)$ and for $\theta = \frac{1}{2} - 2\varepsilon$ with a fixed $0 < \varepsilon < 1/4$ if $\delta \in [1, 3/2)$, we get*

$$\begin{aligned} & \sum_{1 < q \leq N^{\theta}} \frac{1}{\varphi(q)} \left| \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta-n} \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right) \right| \\ & \ll_{f, \beta, \varepsilon, \delta} N^{2-\varepsilon}. \end{aligned}$$

Proof. We begin by considering the case $\delta \in [\frac{3}{2}, 2)$. By the previous explicit formula, the main term and the regularity of the weight,

$$\begin{aligned} & -\frac{1}{\varphi(q)} \sum_{\rho:|\gamma|\leq U} \frac{\Gamma(\rho)}{\zeta'(\rho)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi:|\gamma_\chi|\leq T} \frac{\Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \sum_{k \leq N\beta-1} \Delta_{1/N}^2 \left(f, \frac{k}{N} \right) k^{\rho+\rho_\chi+1} \\ & \ll_{f,\beta} \frac{N^{3-\delta}}{\varphi(q)} \sum_{\rho:|\gamma|\leq U} \left| \frac{\Gamma(\rho)}{\zeta'(\rho)} \right| \sum_{\chi \neq \chi_0} \sum_{\rho_\chi:|\gamma_\chi|\leq T} \left| \frac{\Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \right|. \end{aligned}$$

Now, arguing as in Theorem 22 we have that the double series converges absolutely and it is $\ll \log(q)$, so we have

$$\begin{aligned} & -\frac{1}{\varphi(q)} \sum_{\rho:|\gamma|\leq U} \frac{\Gamma(\rho)}{\zeta'(\rho)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi:|\gamma_\chi|\leq T} \frac{\Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \sum_{k \leq N\beta-1} \Delta_{1/N}^2 \left(f, \frac{k}{N} \right) k^{\rho+\rho_\chi+1} \\ & \ll_{f,\beta} N^{3-\delta} \log(q) \ll_{f,\beta,\delta} N^{3-\delta} \log(N) \end{aligned}$$

Now we consider the error term. Taking $U < T$, $U \asymp T \asymp N$, we obtain

$$\begin{aligned} E_{\max}^* &:= N^{5/2-\delta} \log(qT)^2 + \frac{N^{3-\delta} \log(qNT)^2}{T} \\ &+ \log(q)^3 N^{5/2-\delta+\varepsilon} \log(N)^3 \left[1 + \frac{N \log(qT)^2}{T} \right] \left[1 + \frac{N^{1-\varepsilon}}{U^{1-\varepsilon}} \right] \\ &\ll_{\varepsilon} \log(q)^5 N^{5/2-\delta+\varepsilon} \log(N)^5. \end{aligned}$$

Combining all the pieces and taking $\theta = \delta - 1 - 2\varepsilon$, we get

$$\begin{aligned} & \sum_{1 < q \leq N^{\delta-1-2\varepsilon}} \frac{1}{\varphi(q)} \left| \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta-n} \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right) \right| \\ & \ll_{f,\beta,\varepsilon} N^{2-2\varepsilon} \log(N) + N^{5/2-\delta+\varepsilon} \log(N)^5 \sum_{q \leq N^{\delta-1-2\varepsilon}} \log(q)^5 \ll_{f,\beta,\varepsilon,\delta} N^{2-\varepsilon}. \end{aligned}$$

Now, assume that $\delta \in [1, \frac{3}{2})$. As regards the error term, taking again $U < T$, $U \asymp T \asymp N$, we obtain the same bound

$$E_{\max}^* \ll \log(q)^5 N^{5/2-\delta+\varepsilon} \log(N)^5.$$

So now we focus on the main term, which can be treated in a different way with respect to the previous case, since a direct use of the absolute convergence of the series leads to a too large error in N . Using summation by parts and recalling the property (3.1), which means in this case

$$\Delta_{1/N}^2 \left(f, \frac{k}{N} \right) = \Delta_{1/N} \left(f, \frac{k+1}{N} \right) - \Delta_{1/N} \left(f, \frac{k}{N} \right)$$

we get

$$\begin{aligned} & \sum_{k \leq N\beta-1} \Delta_{1/N}^2 \left(f, \frac{k}{N} \right) k^{\rho+\rho_\chi+1} = \Delta_{1/N}(f, \beta) (N\beta-1)^{\rho+\rho_\chi+1} \\ & - \sum_{k \leq N\beta-1} \Delta_{1/N} \left(f, \frac{k}{N} \right) \left(k^{\rho+\rho_\chi+1} - (k-1)^{\rho+\rho_\chi+1} \right). \end{aligned}$$

We recall that if $f \in \mathcal{C}^1(\mathbb{R})$, then $\Delta_{1/N}(f, x) \ll N^{-1} \log(N)$, by (3.2), so clearly this bound can be used for every fixed $\delta \in [1, \frac{3}{2})$. Hence, we have

$$\Delta_{1/N}(f, \beta) (N\beta - 1)^{\rho + \rho_\chi + 1} \ll_{f, \beta} N \log(N)$$

so it remains to bound the sum. Note that

$$\begin{aligned} & \sum_{k \leq N\beta - 1} \Delta_{1/N}\left(f, \frac{k}{N}\right) \left(k^{\rho + \rho_\chi + 1} - (k-1)^{\rho + \rho_\chi + 1}\right) \\ &= \sum_{k \leq N\beta - 1} \Delta_{1/N}\left(f, \frac{k}{N}\right) (\rho + \rho_\chi + 1) \int_{k-1}^k v^{\rho + \rho_\chi} dv \\ &= \sum_{k \leq N\beta - 1} \Delta_{1/N}\left(f, \frac{k}{N}\right) (\rho + \rho_\chi + 1) \int_0^1 (k-1+t)^{\rho + \rho_\chi} dt \\ &\ll_{f, \beta} \frac{|\rho + \rho_\chi + 1|}{N} \log(N) \sum_{1 < k \leq N\beta - 1} k \ll_{f, \beta} N \log(N) |\rho + \rho_\chi + 1| \end{aligned}$$

so we have to estimate

$$\frac{N \log(N)}{\varphi(q)} \sum_{\rho: |\gamma| \leq U} \frac{1}{|\zeta'(\rho)|} \sum_{\chi \neq \chi_0} \sum_{\rho_\chi: |\gamma_\chi| \leq T} \left| \frac{\Gamma(\rho) \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 1)} \right|$$

and in this case we cannot use the absolute convergence of the series. We start considering $|\gamma + \gamma_\chi| < 1$. Then, recalling that

$$N_\chi(t+1, \chi) - N_\chi(t, \chi) \ll \log(q(|t|+2))$$

which is, essentially, a consequence of (3.4) (see [20], Theorem 10.17), we have, from Proposition 13, that

$$\begin{aligned} & \sum_{\rho: |\gamma| \leq U} \frac{1}{|\zeta'(\rho)|} \sum_{\rho_\chi: |\gamma_\chi| \leq T, |\gamma + \gamma_\chi| < 1} \left| \frac{\Gamma(\rho) \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 1)} \right| \\ &\ll \sum_{\rho: |\gamma| \leq U} \frac{e^{-\pi|\gamma|/2}}{|\zeta'(\rho)|} \sum_{\rho_\chi: |\gamma_\chi| \leq T, |\gamma + \gamma_\chi| < 1} e^{-\pi|\gamma_\chi|/2} \\ &\ll \sum_{\rho: |\gamma| \leq U} \frac{e^{-\pi|\gamma|/2}}{|\zeta'(\rho)|} \# \{\rho_\chi : \gamma_\chi \in (-\gamma - 1, -\gamma + 1)\} \\ &\ll \log(q) \sum_{\rho: |\gamma| \leq U} \frac{e^{-\pi|\gamma|/2}}{|\zeta'(\rho)|} \log(|\gamma|) \ll \log(q). \end{aligned}$$

Now, assume $|\gamma + \gamma_\chi| \geq 1$. Then, again from Proposition 13,

$$\begin{aligned} & \sum_{\rho: |\gamma| \leq U} \frac{1}{|\zeta'(\rho)|} \sum_{\rho_\chi: |\gamma_\chi| \leq T, |\gamma + \gamma_\chi| \geq 1} \left| \frac{\Gamma(\rho) \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 1)} \right| \\ &\ll \sum_{\rho: |\gamma| \leq U} \frac{1}{|\zeta'(\rho)|} \sum_{\rho_\chi: |\gamma_\chi| \leq T, |\gamma + \gamma_\chi| \geq 1} \frac{e^{-\pi(|\gamma| + |\gamma_\chi| - |\gamma + \gamma_\chi|)/2}}{|\gamma + \gamma_\chi|^{3/2}}. \end{aligned}$$

If $\gamma\gamma_\chi > 0$, then it is enough to consider

$$\sum_{\rho: 0 < \gamma \leq U} \frac{1}{|\zeta'(\rho)|} \sum_{\rho_\chi: 0 < \gamma_\chi \leq T, \gamma + \gamma_\chi \geq 1} \frac{1}{(\gamma + \gamma_\chi)^{3/2}}$$

$$\begin{aligned} &\ll \sum_{\rho:0<\gamma\leq U} \frac{1}{|\zeta'(\rho)| \gamma^{3/4}} \sum_{\rho_X:0<\gamma_X\leq T, \gamma+\gamma_X\geq 1} \frac{1}{\gamma_X^{3/4}} \\ &\ll (TU)^{1/4} \log(qT)^2 \log(U)^{1/2} \end{aligned}$$

By (3.3) and Conjecture 19. Assume now $\gamma\gamma_X < 0$ and let $\gamma > 0, \gamma_X < 0$. We have

$$\begin{aligned} &\sum_{\rho:0<\gamma\leq U} \frac{1}{|\zeta'(\rho)|} \sum_{\rho_X:0<\gamma_X\leq T, \gamma-\gamma_X\geq 1} \frac{e^{-\pi\gamma_X}}{(\gamma-\gamma_X)^{3/2}} \\ &+ \sum_{\rho:0<\gamma\leq U} \frac{e^{-\pi\gamma}}{|\zeta'(\rho)|} \sum_{\rho_X:0<\gamma_X\leq T, \gamma-\gamma_X\leq -1} \frac{1}{(\gamma_X-\gamma)^{3/2}} \end{aligned}$$

then note that

$$\gamma - \gamma_X \geq 1 \Rightarrow \gamma_X \leq \gamma - 1 \leq U - 1 < T$$

hence we can consider

$$\sum_{\rho_X:0<\gamma_X\leq T, \gamma-\gamma_X\leq -1} \frac{e^{-\pi\gamma_X}}{(\gamma-\gamma_X)^{3/2}} = \sum_{\rho_X:0<\gamma_X\leq \gamma-1} \frac{e^{-\pi\gamma_X}}{(\gamma-\gamma_X)^{3/2}}$$

now if $\gamma_X < \gamma/2$ then

$$\sum_{\rho_X:0<\gamma_X\leq \gamma/2} \frac{e^{-\pi\gamma_X}}{(\gamma-\gamma_X)^{3/2}} \ll \frac{\log(q)}{\gamma^{3/2}}$$

and if $\gamma/2 \leq \gamma_X < \gamma - 1$ then

$$\sum_{\rho_X:\gamma/2 < \gamma_X \leq \gamma-1} \frac{e^{-\pi\gamma_X}}{(\gamma-\gamma_X)^{3/2}} \ll e^{-\pi\gamma/2} \gamma \log(q\gamma)$$

then

$$\begin{aligned} &\sum_{\rho:0<\gamma\leq U} \frac{1}{|\zeta'(\rho)|} \sum_{\rho_X:0<\gamma_X\leq T, \gamma-\gamma_X\geq 1} \frac{e^{-\pi\gamma_X}}{(\gamma-\gamma_X)^{3/2}} \\ &\ll \log(q) \left[\sum_{\rho:0<\gamma\leq U} \frac{1}{|\zeta'(\rho)| \gamma^{3/2}} + \sum_{\rho:0<\gamma\leq U} \frac{\gamma \log(\gamma)}{|\zeta'(\rho)|} \right] \ll \log(q) \end{aligned}$$

due to the absolute convergence of the series. For the second sum we clearly have

$$\begin{aligned} &\sum_{\rho_X:0<\gamma_X\leq T, \gamma-\gamma_X\leq -1} \frac{1}{(\gamma_X-\gamma)^{3/2}} = \sum_{\rho_X:\gamma+1\leq \gamma_X\leq T} \frac{1}{(\gamma_X-\gamma)^{3/2}} \\ &\ll \sum_{m\in[1, T-\gamma]\cap\mathbb{N}} \sum_{\gamma_X\in[\gamma+m, \gamma+m+1)} \frac{1}{(\gamma_X-\gamma)^{3/2}} \\ &\ll \sum_{m\geq 1} \frac{\log(q(\gamma+m+2))}{m^{3/2}} \ll \log(q) + \log(\gamma) \end{aligned}$$

hence

$$\sum_{\rho:0<\gamma\leq U} \frac{e^{-\pi\gamma}}{|\zeta'(\rho)|} \sum_{\rho_X:0<\gamma_X\leq T, \gamma-\gamma_X\leq -1} \frac{1}{(\gamma_X-\gamma)^{3/2}} \ll \log(q).$$

It remains to control what happens if $\gamma_\chi = 0$, which is a case that we cannot exclude. We can argue as in Theorem 15, and observe, by Stirling formula (3.5), that

$$\begin{aligned} \sum_{\rho: |\gamma| \leq U} \frac{1}{|\zeta'(\rho)|} \sum_{\rho_\chi: \gamma_\chi = 0} \left| \frac{\Gamma(\rho) \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 1)} \right| &\ll N(4, \chi) \sum_{\rho: |\gamma| \leq U} \frac{|\Gamma(\rho)|}{|\zeta'(\rho)| |\Gamma(\rho + \frac{3}{2})|} \\ &\ll \log(q) \sum_{\rho} \frac{1}{|\zeta'(\rho)| |\gamma|^{3/2}} \end{aligned}$$

which is convergent, by Conjecture 19.

Finally, we obtain

$$\begin{aligned} \frac{N \log(N)}{\varphi(q)} \sum_{\rho: |\gamma| \leq U} \frac{1}{|\zeta'(\rho)|} \sum_{\chi \neq \chi_0} \sum_{\rho_\chi: |\gamma_\chi| \leq T} \left| \frac{\Gamma(\rho) \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 1)} \right| \\ \ll N \log(N) \log(q) (TU)^{1/4} \log(qT)^2 \log(U)^{1/2} \end{aligned}$$

and since $U \asymp T \asymp N$ we can conclude

$$\begin{aligned} \frac{N \log(N)}{\varphi(q)} \sum_{\rho: |\gamma| \leq U} \frac{1}{|\zeta'(\rho)|} \sum_{\chi \neq \chi_0} \sum_{\rho_\chi: |\gamma_\chi| \leq T} \left| \frac{\Gamma(\rho) \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 1)} \right| \\ \ll N^{3/2} \log(N)^{9/4} \log(q)^2. \end{aligned}$$

Again, combining all the pieces, we get, taking $\theta = 1/2 - 2\varepsilon$, that

$$\begin{aligned} \sum_{1 < q \leq N^\theta} \frac{1}{\varphi(q)} \left| \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta - n} \Lambda(n) \chi(n) \mu(m) f\left(\frac{n+m}{N}\right) \right| \\ \ll_{f, \beta} N^{3/2} \log(N)^{9/4} \sum_{1 < q \leq N^{1/2-2\varepsilon}} \log(q)^2 + N^{5/2-\delta} \log(N)^5 \sum_{1 < q \leq N^{1/2-2\varepsilon}} \log(q)^5 \\ \ll_{f, \beta, \varepsilon} N^{3/2} \log(N)^5 \sum_{q \leq N^{1/2-2\varepsilon}} \log(q)^5 \ll_{f, \beta, \varepsilon} N^{2-\varepsilon} \end{aligned}$$

and the claim follows. $\square$

5. WEIGHTED AVERAGES OF DIAGONAL TWISTED EH PROBLEM WITH $g(n) = \mu(n) \log(n)$

In this section we will use the previous results to show that, also in this case, we are able to show that an analogous weighted average for the diagonal twisted EH problem with $g(n) = \mu(n) \log(n)$, that is the average

$$\sum_{N\alpha < n \leq N\beta} \sum_{m \leq N\beta - n} \Lambda(n) \chi(n) \mu(m) \log(m) f\left(\frac{n+m}{N}\right)$$

is consistent with Conjecture 6. The proofs are similar, so we omit some repetitive details and we focus on the main difference between these two cases. We start again considering the weight in the Sobolev space $W^{2,1}$.

Theorem 25. *Assume the hypotheses of Theorem 21. Then*

$$\frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta - n} \Lambda(n) \chi(n) \mu(m) \log(m) f\left(\frac{n+m}{N}\right)$$

$$\begin{aligned}
&= -\frac{1}{\varphi(q)} \sum_{\rho: |\gamma| \leq U} \frac{N^\rho \Gamma(\rho)}{\zeta'(\rho)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{N^{\rho_\chi} \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \int_0^\beta f''(w) \mathcal{H}(N, w, \rho, \rho_\chi) w^{\rho + \rho_\chi + 1} dw \\
&\quad + O_\varepsilon \left(\frac{1}{N} \int_0^\beta |f''(w)| [\mathcal{E}_1^*(N, w, q, U, T) + \mathcal{E}_2^*(N, w, q, U, T)] dw \right)
\end{aligned}$$

where

$$(5.1) \quad \mathcal{H}(N, w, \rho, \rho_\chi) := \log(Nw) + \psi^{(0)}(\rho + 1) - \psi^{(0)}(\rho_\chi + \rho + 2) - \frac{1}{\rho}$$

and $\psi^{(0)}(z)$ is the Digamma function,

$$\begin{aligned}
(5.2) \quad \mathcal{E}_1^*(N, w, q, U, T) &:= \varphi(q) N^{3/2+\varepsilon} \log(q)^3 \log(N)^3 \left(1 + w^{3/2+\varepsilon}\right) \left(1 + |\log(w)|^3\right) \\
&\quad \times \left[1 + \frac{N(1+w) \log(qT)^2}{T} + \frac{N^{1-\varepsilon} (w^{1-\varepsilon} + 1)}{U^{1-\varepsilon}} \right],
\end{aligned}$$

$$\begin{aligned}
(5.3) \quad \mathcal{E}_2^*(N, w, q, U, T) &= \varphi(q) N^{1+\varepsilon} \log(q) \log(N)^3 \left(1 + |\log(w)|^3\right) \\
&\quad \times (1 + w^{1+\varepsilon}) \left[1 + \frac{N \log(qT)^2}{T} \right] \left[1 + \frac{N^{1-1\varepsilon}}{U^{1-1\varepsilon}} \right].
\end{aligned}$$

and the implicit constant depends only on ε .

Proof. From Theorem 8 we have

$$\begin{aligned}
(5.4) \quad &\frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta - n} \Lambda(n) \chi(n) \mu(m) \log(m) f\left(\frac{n+m}{N}\right) \\
&= \frac{1}{N\varphi(q)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \int_0^\beta f''(w) \int_0^{Nw} \widetilde{M}(s) \psi(Nw - s, \chi) ds.
\end{aligned}$$

We made a dissection like (4.2), (4.3) and (4.4). Recalling the definition of Digamma function

$$\psi^{(0)}(z) := \frac{\Gamma'(z)}{\Gamma(z)}, z \neq 0, -1, -2, \dots$$

(see [23], 5.2.2) and the representation

$$\psi^{(0)}(a) - \psi^{(0)}(a+b) = \frac{1}{B(a, b)} \int_0^1 \log(s) s^{a-1} (1-s)^{b-1} ds, \operatorname{Re}(a) > 0, \operatorname{Re}(b) > 0$$

which is a simple consequence of the differentiation under the integral sign, we get that the main term of (5.4) is

$$-\frac{1}{\varphi(q)} \sum_{\rho: |\gamma| \leq U} \frac{N^\rho \Gamma(\rho)}{\zeta'(\rho)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{N^{\rho_\chi} \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \int_0^\beta f''(w) \mathcal{H}(N, w, \rho, \rho_\chi) w^{\rho + \rho_\chi + 1} dw$$

where $\mathcal{H}(N, w, \rho, \rho_\chi)$ is (5.1). So it remains to evaluate the error term. Now, by (3.14) and arguing as in Theorem 21, we may deduce that the uniform bound for $0 < x \leq CN$, $C > 0$

$$(5.5) \quad \widetilde{M}(x) \ll \log(N) N^{1/2+\varepsilon}$$

hence defining

$$I_1^* = I_1^*(w, N, q, T) := \int_0^{Nw} \widetilde{M}(s) \sum_{\chi \neq \chi_0} \overline{\chi}(N) R_{1,\chi}(Nw - s, T, q) ds$$

and by (5.5) and following the same calculations of I_1 in Theorem 21, we get

$$(5.6) \quad I_1^* \ll \varphi(q) \log(q) N^{3/2+\varepsilon} \log(N)^3 w^{3/2+\varepsilon} \left(1 + |\log(w)|^3\right) \left[1 + \frac{N(1+w) \log(qT)^2}{T}\right].$$

Then we consider

$$I_2^* = I_2^*(w, N, q, U) := \int_0^{Nw} \widetilde{R}_2(s, U) \sum_{\chi \neq \chi_0} \overline{\chi}(N) \psi(Nw - s, \chi) ds$$

where $\widetilde{R}(s, U)$ is the error term in (3.14), that is

$$\widetilde{M}(s) = \sum_{\rho: |\gamma| \leq U} \frac{x^\rho}{\zeta'(\rho)\rho} \left(\log(x) - \frac{1}{\rho} \right) + \widetilde{R}_2(s, U).$$

Note that, compared to the estimate (4.6) and (4.7) we have only the additional term

$$\begin{aligned} & \varphi(q) N^{1/2} w^{1/2} |\log(Nw)| |\log(qNw)| \int_0^{yw} |\log(s)| ds \\ & \ll \varphi(q) N^{3/2} w^{3/2} \log(Nw)^2 |\log(qNw)| \end{aligned}$$

and this produces the bound

$$(5.7) \quad I_2^* \ll \varphi(q) N^{3/2+\varepsilon} \log(q)^3 \log(N)^3 \times \left[w^{3/2} (1 + w^\varepsilon) \left(1 + |\log(w)|^3\right) + \frac{N^{1-\varepsilon} \left(1 + |\log(w)|^3\right) w^{1/2+\varepsilon} (w^{2-\varepsilon} + 1)}{U^{1-\varepsilon}} \right].$$

Finally, we evaluate

$$I_3^* = I_3^*(w, N, q, U) := \int_0^{Nw} \widetilde{R}_2(s, U) \sum_{\chi \neq \chi_0} \overline{\chi}(N) R_{1,\chi}(Nw - s, T, q) ds.$$

Arguing as in I_3 of Theorem 21, we observe that the only part we need to modified is

$$\begin{aligned} & \int_0^{Nw} \widetilde{R}_2(s, U)^2 ds \ll \int_0^{Nw} \left[1 + s^{2\varepsilon} + \log(s)^2 + \left[\frac{(s^2 + s^{2\varepsilon}) (|\log(s)|^2 + 1)}{U^{2-2\varepsilon}} \right] \right] ds \\ & \ll N^{1+2\varepsilon} \log(N)^2 \left(1 + \log(w)^2\right) (1 + w^{1+2\varepsilon}) \left[1 + \frac{N^{2-2\varepsilon} (1 + w^{2-2\varepsilon}) (1 + \log(w)^2)}{U^{2-2\varepsilon}} \right]. \end{aligned}$$

So

$$(5.8) \quad I_3^* \ll \varphi(q) N^{1+\varepsilon} \log(q) \log(N)^3 \left(1 + |\log(w)|^3\right) (1 + w^{1+\varepsilon})$$

$$(5.9) \quad \times \left[1 + \frac{N \log(qT)^2}{T} \right] \left[1 + \frac{N^{1-1\varepsilon}}{U^{1-1\varepsilon}} \right].$$

Again, for simplicity, we observe that we can use the common bound

$$(5.10) \quad I_1^*, I_2^* \ll \varphi(q) N^{3/2+\varepsilon} \log(q)^3 \log(N)^3 \left(1 + w^{3/2+\varepsilon}\right) \left(1 + |\log(w)|^3\right)$$

$$(5.11) \quad \times \left[1 + \frac{N(1+w) \log(qT)^2}{T} + \frac{N^{1-\varepsilon}(w^{1-\varepsilon} + 1)}{U^{1-\varepsilon}} \right]$$

and the result follows. $\square$

So now we are able to prove the bound for the diagonal EH with $g(n) = \mu(n) \log(n)$ in the case that the weight is in Sobolev $W^{2,1}$.

Theorem 26. *Assume all the hypotheses of Theorem 21 and fix $0 < \varepsilon < 1/2$ and $\theta = 1 - 2\varepsilon$. Then, the following weighted average of diagonal twisted EH holds*

$$\sum_{\substack{1 < q \leq N^{1-2\varepsilon} \\ (N,q)=1}} \frac{1}{\varphi(q)} \left| \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta-n} \Lambda(n) \chi(n) \mu(m) \log(m) f\left(\frac{n+m}{N}\right) \right| \\ \ll_{\varepsilon} N^{2-\varepsilon} E_1(f'')$$

where

$$E_1(f'') := \int_0^\beta |f''(w)| \left(1 + |\log(w)|^3\right) \left(1 + w^{5/2+\varepsilon}\right) dw.$$

and the implicit constant depends only on ε .

Proof. From Theorem 25 we know that we need to bound the explicit formula. We start from the main term. Recalling the asymptotic formula

$$(5.12) \quad \psi^{(0)}(z) \sim \log(z) - \frac{1}{2z} - \sum_{k \geq 0} \frac{B_{2k}}{2k z^{2k}}$$

as $|z| \rightarrow +\infty$, $|\arg(z)| \leq \pi - \epsilon$, $\epsilon > 0$ (see [23], 5.11.2) where $\log(z)$ is taken with its principal branch, we obtain

$$\psi^{(0)}(\rho + 1) - \psi^{(0)}(\rho_\chi + \rho + 2) - \frac{1}{\rho} \\ \ll \log(|\gamma|) + \log(1 + |\gamma + \gamma_\chi|) + \frac{1}{|\rho|} \\ \ll \log(T + U)$$

hence

$$\frac{1}{\varphi(q)} \sum_{\rho: |\gamma| \leq U} \frac{N^\rho \Gamma(\rho)}{\zeta'(\rho)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{N^{\rho_\chi} \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \int_0^\beta f''(w) \mathcal{H}(N, w, \rho, \rho_\chi) w^{\rho + \rho_\chi + 1} dw \\ \ll \frac{N \log(N) \log(T + U)}{\varphi(q)} \sum_{\rho: |\gamma| \leq U} \left| \frac{\Gamma(\rho)}{\zeta'(\rho)} \right| \sum_{\chi \neq \chi_0} \sum_{\rho_\chi: |\gamma_\chi| \leq T} \left| \frac{\Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \right| \\ \times \int_0^\beta |f''(w)| (1 + |\log(w)|) w^2 dw$$

hence if we take $T \asymp U \asymp N$ we get the bound

$$N \log(N)^2 \max_{\chi \bmod q} \sum_{\rho: |\gamma| \leq U} \left| \frac{\Gamma(\rho)}{\zeta'(\rho)} \right| \sum_{\rho_\chi: |\gamma_\chi| \leq T} \left| \frac{\Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \right|$$

$$\times \int_0^\beta |f''(w)| (1 + |\log(w)|) w^2 dw$$

and arguing as in (4.8) we get that the double series is absolutely convergent and bounded by $\log(q)$. Hence the bound for the main term is

$$N \log(N)^2 \log(q) \int_0^\beta |f''(w)| (1 + |\log(w)|) w^2 dw.$$

Now let us focus on the error term. By (5.2), (5.3), from the choice $T \asymp U \asymp N$ and trivial manipulations we get

$$\mathcal{E}_1^*(N, w, q, U, T) \ll \varphi(q) N^{3/2+\varepsilon} \log(q)^5 \log(N)^5 \left(1 + w^{5/2+\varepsilon}\right) \left(1 + |\log(w)|^3\right)$$

$$\mathcal{E}_2^*(N, w, q, U, T) \ll \varphi(q) N^{1+\varepsilon} \log(q)^2 \log(N)^5 \left(1 + |\log(w)|^3\right) (1 + w^{1+\varepsilon})$$

hence, putting together all the pieces and taking the sum $\sum_{\substack{q \leq N^\theta \\ (N, q)=1}}$ we get

$$\sum_{\substack{q \leq N^\theta \\ (N, q)=1}} \frac{1}{\varphi(q)} \left| \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta-n} \Lambda(n) \chi(n) \mu(m) \log(m) f\left(\frac{n+m}{N}\right) \right|$$

$$\ll_\varepsilon N^{2-\varepsilon} E_1(f'')$$

with $\theta = 1 - 2\varepsilon$ and

$$E_1(f'') := \int_0^\beta |f''(w)| \left(1 + |\log(w)|^3\right) \left(1 + w^{5/2+\varepsilon}\right) dw.$$

□

Now we consider the case of $f \in \mathcal{C}^\delta(\mathbb{R})$. Also in this case, we start with an explicit formula for the average.

Theorem 27. *Assume all the hypotheses of Theorem 23. Then*

$$\begin{aligned} & \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta-n} \Lambda(n) \chi(n) \mu(m) \log(m) f\left(\frac{n+m}{N}\right) \\ &= -\frac{1}{\varphi(q)} \sum_{\rho: |\gamma| \leq U} \frac{\Gamma(\rho)}{\zeta'(\rho)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{\Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \\ & \quad \times \sum_{k \leq \beta N-1} \Delta_{1/N}^2\left(f, \frac{k}{N}\right) \mathcal{H}\left(N, \frac{k}{N}, \rho, \rho_\chi\right) k^{\rho+\rho_\chi+1} \\ &+ O_{f, \beta, \varepsilon} \left(N^{5/2-\delta} \log(N) \log(qT)^2 + \frac{N^{3-\delta} \log(N) \log(qNT)^2}{T} \right. \\ & \left. + \log(q)^3 \varphi(q) N^{5/2-\delta+\varepsilon} \log(N)^3 \left[1 + \frac{N \log(qT)^2}{T} \right] \left[1 + \frac{N^{1-\varepsilon}}{U^{1-\varepsilon}} \right] \right). \end{aligned}$$

Proof. Arguing as in Theorem 23 we have to deal with

$$\begin{aligned}
& \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta - n} \Lambda(n) \chi(n) \mu(m) \log(m) f\left(\frac{n+m}{N}\right) \\
&= \frac{1}{\varphi(q)} \sum_{k \leq N\beta-1} \Delta_{1/N}^2\left(f, \frac{k}{N}\right) \sum_{\chi \neq \chi_0} \bar{\chi}(N) \int_0^k \psi(s, \chi) \widetilde{M}(k-s) ds \\
&+ O_{f,\beta} \left(\frac{N^{-\delta}}{\varphi(q)} \sum_{k \leq N\beta-1} \sum_{\chi \neq \chi_0} \left| \sum_{u=0}^k \psi(\chi, u) \mu(k-u) \log(k-u) \right| \right).
\end{aligned}$$

By the explicit formula in Theorem 25 we have

$$\begin{aligned}
& \frac{1}{\varphi(q)} \sum_{k \leq N\beta-1} \Delta_{1/N}^2\left(f, \frac{k}{N}\right) \sum_{\chi \neq \chi_0} \bar{\chi}(N) \int_0^k \psi(s, \chi) \widetilde{M}(k-s) ds \\
&= \frac{1}{\varphi(q)} \sum_{\rho: |\gamma| \leq U} \frac{\Gamma(\rho)}{\zeta'(\rho)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{\Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \\
&\quad \times \sum_{k \leq \beta N-1} \Delta_{1/N}^2\left(f, \frac{k}{N}\right) \mathcal{H}\left(N, \frac{k}{N}, \rho, \rho_\chi\right) k^{\rho+\rho_\chi+1} \\
&+ O_f \left(\frac{N^{-\delta}}{\varphi(q)} \sum_{k \leq N\beta-1} [I_1^*(k/N, N, q, T) + I_2^*(k/N, N, q, U) + I_3^*(k/N, N, q, T, U)] \right) \\
&+ O_{f,\beta} \left(\frac{N^{-\delta}}{\varphi(q)} \sum_{k \leq N\beta-1} \sum_{\chi \neq \chi_0} \left| \sum_{u=0}^k \psi(u, \chi) \mu(k-u) \log(k-u) \right| \right).
\end{aligned}$$

We start to analyze the first error term. From the same calculations of (5.10), (5.11), and recalling that the convolution now is from 0 to k instead of 0 to Nw , we obtain the bound

$$\begin{aligned}
& \log(q)^3 \varphi(q) N^{-\delta} \sum_{k \leq N\beta-1} k^{3/2+\varepsilon} \log(k)^3 \left[1 + \frac{k \log(qT)^2}{T} + \frac{k^{1-\varepsilon}}{U^{1-\varepsilon}} \right] \\
&\ll_{\varepsilon,\beta} \log(q)^3 \varphi(q) N^{5/2-\delta+\varepsilon} \log(N)^3 \left[1 + \frac{N \log(qT)^2}{T} + \frac{N^{1-\varepsilon}}{U^{1-\varepsilon}} \right]
\end{aligned}$$

and in the same spirit, considering (5.8) and (5.9), we obtain the bound

$$\begin{aligned}
& \log(q) \varphi(q) N^{-\delta} \sum_{k \leq N\beta-1} k^{1+\varepsilon} \log(k)^3 \left[1 + \frac{k \log(qT)^2}{T} \right] \left[1 + \frac{k^{1-\varepsilon}}{U^{1-1\varepsilon}} \right] \\
&\ll_{\varepsilon,\beta} \log(q) \varphi(q) N^{2+\varepsilon-\delta} \log(N)^3 \left[1 + \frac{N \log(qT)^2}{T} \right] \left[1 + \frac{N^{1-\varepsilon}}{U^{1-1\varepsilon}} \right].
\end{aligned}$$

Now, we consider the second error term. Again by the explicit formula in Theorem 17 we get

$$\frac{N^{-\delta}}{\varphi(q)} \sum_{k \leq N\beta-1} \sum_{\chi \neq \chi_0} \left| \sum_{u=0}^k \psi(u, \chi) \mu(k-u) \log(k-u) \right|$$

$$\begin{aligned}
&= \frac{N^{-\delta}}{\varphi(q)} \sum_{k \leq N\beta-1} \sum_{\chi \neq \chi_0} \left| \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{1}{\rho_\chi} \sum_{u=1}^{k-1} u^{\rho_\chi} \mu(k-u) \log(k-u) \right| \\
&+ O_\beta \left(N^{2-\delta} \log(N)^2 + N^{-\delta} \log(N) \sum_{k \leq N\beta-1} \sum_{u=1}^{k-1} \frac{u \log(quT)^2}{T} \right).
\end{aligned}$$

Again we have

$$N^{-\delta} \log(N) \sum_{k \leq N\beta-1} \sum_{u=1}^{k-1} \frac{u \log(quT)^2}{T} \ll \frac{N^{3-\delta} \log(N) \log(qNT)^2}{T}$$

and

$$\begin{aligned}
&\frac{N^{-\delta}}{\varphi(q)} \sum_{k \leq N\beta-1} \sum_{\chi \neq \chi_0} \left| \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{1}{\rho_\chi} \sum_{u=1}^{k-1} u^{\rho_\chi} \mu(k-u) \log(k-u) \right| \\
&\ll \frac{N^{-\delta} \log(N)}{\varphi(q)} \sum_{k \leq N\beta-1} k^{3/2} \sum_{\chi \neq \chi_0} \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{1}{|\rho_\chi|} \ll N^{5/2-\delta} \log(N) \log(qT)^2
\end{aligned}$$

and the thesis follows combining all the pieces. $\square$

Finally, we can prove the bound for the diagonal EH problem with $f \in \mathcal{C}^\delta(\mathbb{R})$.

Theorem 28. *Assume all the hypotheses of Theorem 23 and fix $\theta = \delta - 1 - 2\varepsilon$ if $\delta \in [\frac{3}{2}, 2)$ with $0 < \varepsilon < (\delta - 1)/2$ and $\theta = \frac{1}{2} - 2\varepsilon$ if $\delta \in [1, 3/2)$ with $0 < \varepsilon < 1/4$. Then*

$$\begin{aligned}
&\sum_{\substack{1 < q \leq N^\theta \\ (N, q) = 1}} \frac{1}{\varphi(q)} \left| \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta-n} \Lambda(n) \chi(n) \mu(m) \log(m) f\left(\frac{n+m}{N}\right) \right| \\
&\ll_{f, \beta, \varepsilon, \delta} N^{2-\varepsilon}.
\end{aligned}$$

Proof. We start again considering the case $\delta \in [\frac{3}{2}, 2)$. By the explicit formula in Theorem 27 and (5.12) we get

$$\begin{aligned}
&\frac{1}{\varphi(q)} \sum_{\rho: |\gamma| \leq U} \frac{\Gamma(\rho)}{\zeta'(\rho)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{\Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \\
&\times \sum_{k \leq N\beta-1} \Delta_{1/N}^2 \left(f, \frac{k}{N}\right) \mathcal{H}\left(N, \frac{k}{N}, \rho, \rho_\chi\right) k^{\rho+\rho_\chi+1} \\
&\ll_{f, \beta} \frac{N^{3-\delta} \log(T+U)}{\varphi(q)} \sum_{\rho: |\gamma| \leq U} \left| \frac{\Gamma(\rho)}{\zeta'(\rho)} \right| \sum_{\chi \neq \chi_0} \sum_{\rho_\chi: |\gamma_\chi| \leq T} \left| \frac{\Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \right|
\end{aligned}$$

then, taking $T \asymp U \asymp N$ and arguing as in Theorem 22 for the analysis of the double series, we obtain the bound

$$\ll_{f, \beta, \delta} \frac{N^{3-\delta} \log(N) \log(q)}{\varphi(q)} \ll_{f, \beta, \delta} \frac{N^{3-\delta} \log(N)^2}{\varphi(q)}.$$

For what concerning the error term, with the choice $T \asymp U \asymp N$, we obtain

$$E_{\max}^{**} := N^{5/2-\delta+\varepsilon} \log(N) \log(qT)^2 + \frac{N^{3-\delta} \log(N) \log(qNT)^2}{T}$$

$$\begin{aligned}
& + \log(q)^3 N^{5/2-\delta+\varepsilon} \log(N)^3 \left[1 + \frac{N \log(qT)^2}{T} \right] \left[1 + \frac{N^{1-\varepsilon}}{U^{1-1\varepsilon}} \right] \\
& \ll_{\varepsilon, \theta} N^{5/2-\delta+\varepsilon} \log(N)^8
\end{aligned}$$

hence, taking $\theta = \delta - 1 - 2\varepsilon$, we obtain

$$\begin{aligned}
& \sum_{1 < q \leq N^{\delta-1-2\varepsilon}} \frac{1}{\varphi(q)} \left| \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta-n} \Lambda(n) \chi(n) \mu(m) \log(m) f\left(\frac{n+m}{N}\right) \right| \\
& \ll_{f, \beta, \varepsilon, \delta} N^{2-2\varepsilon} \log(N)^2 + N^{3/2-2\varepsilon} \log(N)^8 \ll_{f, \beta, \varepsilon, \delta} N^{2-\varepsilon}.
\end{aligned}$$

Now we focus on the case $\delta \in [1, \frac{3}{2})$. Assuming again that $U \asymp T \asymp N$ the bound of the error term is the same of the previous part, so

$$E_{\max}^{**} \ll_{\varepsilon} N^{5/2-\delta+\varepsilon} \log(N)^8$$

So now we focus on the main term. Clearly

$$\mathcal{H}\left(N, \frac{k}{N}, \rho, \rho_{\chi}\right) = \log(k) + \psi^{(0)}(\rho+1) - \psi^{(0)}(\rho_{\chi} + \rho + 2) - \frac{1}{\rho}$$

hence the significant difference here respect the previous results is the factor $\log(k)$ in the sum. Using summation by parts and property (3.1) we observe that

$$\begin{aligned}
& \sum_{k \leq N\beta-1} \Delta_{1/N}^2\left(f, \frac{k}{N}\right) \log(k) k^{\rho+\rho_{\chi}+1} = \Delta_{1/N}(f, \beta) \log(N\beta) (N\beta-1)^{\rho+\rho_{\chi}+1} \\
& - \sum_{k \leq N\beta-1} \Delta_{1/N}\left(f, \frac{k}{N}\right) \log(k) \left(k^{\rho+\rho_{\chi}+1} - (k-1)^{\rho+\rho_{\chi}+1}\right).
\end{aligned}$$

From (3.2) we have

$$\Delta_{1/N}(f, \beta) \log(N\beta) (N\beta-1)^{\rho+\rho_{\chi}+1} \ll_{\beta} N \log(N)^2$$

and

$$\begin{aligned}
& \sum_{k \leq N\beta-1} \Delta_{1/N}\left(f, \frac{k}{N}\right) \log(k) \left(k^{\rho+\rho_{\chi}+1} - (k-1)^{\rho+\rho_{\chi}+1}\right) \\
& \ll_{f, \beta} N \log(N)^2 |\rho + \rho_{\chi} + 1|
\end{aligned}$$

using the same ideas of Theorem 9. So, since the part $\psi^{(0)}(\rho+1) - \psi^{(0)}(\rho_{\chi} + \rho + 2) - \frac{1}{\rho}$ produces a bound $\log(T+U) \ll \log(N)$, for bounding the main term it is just enough to consider

$$\begin{aligned}
& \frac{N \log(N)^2}{\varphi(q)} \sum_{\rho: |\gamma| \leq N} \frac{1}{|\zeta'(\rho)|} \sum_{\chi \neq \chi_0} \sum_{\rho_{\chi}: |\gamma_{\chi}| \leq N} \left| \frac{\Gamma(\rho) \Gamma(\rho_{\chi})}{\Gamma(\rho_{\chi} + \rho + 1)} \right| \\
& \ll N^{3/2} \log(N)^2 \log(q) \log(qN) \log(N)^{1/4} \\
& \ll N^{3/2} \log(N)^{13/4} \log(q).
\end{aligned}$$

So, combining the pieces and taking $\theta = 1/2 - 2\varepsilon$, we finally obtain

$$\begin{aligned}
& \sum_{1 < q \leq N^{\theta}} \frac{1}{\varphi(q)} \left| \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N\beta} \sum_{m \leq N\beta-n} \Lambda(n) \chi(n) \mu(m) \log(m) f\left(\frac{n+m}{N}\right) \right| \\
& \ll_{f, \beta} N^{3/2} \log(N)^{17/4} \sum_{q \leq N^{1/2-2\varepsilon}} \log(q) + N^{3-\delta-2\varepsilon} \log(N)^8
\end{aligned}$$

$$\ll_{f,\beta,\varepsilon} N^{2-\varepsilon}$$

and so the thesis. $\square$

6. EXAMPLES

In this section we propose some interesting examples that can be easily deduced by the previous theorems, and that illustrate both the Sobolev and the Hölder–Zygmund frameworks through standard compactly supported weights.

We start with the classical Cesàro–Riesz weights, that is, the function

$$f_{CR,k}(x) := \begin{cases} (1-x)^k, & x \in [0, 1] \\ 0, & \text{otherwise} \end{cases}$$

where $k \in \mathbb{R}_0^+$. If we assume that $k > 1$, it is simple to observe that $f_{CR,k}(x)$ verifies the hypotheses of Theorem 8, and so, making a simple change of variable, we have immediately the following results.

Theorem 29. *Fix $0 < \varepsilon < 1/2$. Assume GRH and Conjecture 19. Let $N \geq 4$ and $k > 1$, $q \in \mathbb{N}$, $1 < q < N^\theta$, $0 < \theta < 1$. Then*

$$\begin{aligned} & \frac{1}{\varphi(q) \Gamma(k+1)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N} \left(1 - \frac{n}{N}\right)^k \sum_{m \leq n} \Lambda(m) \chi(m) \mu(n-m) \\ &= -\frac{1}{\varphi(q)} \sum_{\rho} \frac{N^\rho \Gamma(\rho)}{\zeta'(\rho)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi} \frac{N^{\rho_\chi} \Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + k + 1)} \\ &+ O_\varepsilon \left(\frac{k(k-1) N^{1/2+\varepsilon} \log(q)^3 \log(N)^3}{\Gamma(k+1)} \int_0^1 (1-w)^{k-2} \left(1 + w^{3/2+\varepsilon}\right) \left(1 - \log(w)^3\right) dw \right) \end{aligned}$$

and the implicit constant depends only on ε .

Note that by Theorem 15 the double series is absolutely convergent, and this is the reason of the form of the previous theorem. Then, we can deduce the following theorem.

Theorem 30. *Assume all the hypotheses of the previous theorem and fix $0 < \varepsilon < 1/2$ and $\theta = 1 - 2\varepsilon$. Then, the following weighted average of diagonal twisted EH holds*

$$\sum_{1 < q \leq N^{1-2\varepsilon}} \frac{1}{\varphi(q) \Gamma(k+1)} \left| \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N} \left(1 - \frac{n}{N}\right)^k \sum_{m \leq n} \Lambda(m) \chi(m) \mu(n-m) \right| \ll_\varepsilon N^{2-\varepsilon} E(f'')$$

where

$$E(f'') := \frac{k(k-1)}{\Gamma(k+1)} \int_0^1 (1-w)^{k-2} \left(1 - \log(w)^3\right) (1+w^2) dw.$$

and the implicit constant depends only on ε .

Actually, since the error terms of the previous theorem depend explicitly on k , the main result can be extended also to $k = 1$. Now, take

$$f_{Zyg}(x) := \begin{cases} (1-x) \sin(\log(1-x)), & x \in [0, 1] \\ 0, & \text{otherwise;} \end{cases}$$

now, it is clear that, taking for instance $\alpha = 0$ and $\beta = 1 + \varepsilon$ with $\varepsilon \in (0, 1/2)$, then $f_{Zyg}(x)$ does not fit the hypotheses of Theorem 8 but this function is in $\mathcal{C}^1(\mathbb{R})$. Indeed,

$$f'_{Zyg}(x) = -\sin(\log(1-x)) - \cos(\log(1-x))$$

hence

$$|f'_{Zyg}(x)| \leq 2$$

and, by continuity, $f_{Zyg}(0) = f_{Zyg}(1) = 0$, so $f_{Zyg}(x) \in \text{Lip}(\mathbb{R}) \subset \mathcal{C}^1(\mathbb{R})$.

Observe that this function is a variant of the Cesàro-Riesz weight of order 1. Then, for this function, we can prove what follows.

Theorem 31. *Fix $0 < \varepsilon < 1/4$. Assume GRH, let $N \geq 4$, $T, U \geq 4$, $q \in \mathbb{N}$, $1 < q < N^\theta$, $0 < \theta < 1$. Assume also the Conjecture 19. Then*

$$\begin{aligned} & \frac{1}{\varphi(q)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N} \sum_{m \leq N-n} \Lambda(n) \chi(n) \mu(m) f_{Zyg}\left(\frac{n+m}{N}\right) \\ &= -\frac{1}{\varphi(q)} \sum_{\rho: |\gamma| \leq U} \frac{\Gamma(\rho)}{\zeta'(\rho)} \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{\rho_\chi: |\gamma_\chi| \leq T} \frac{\Gamma(\rho_\chi)}{\Gamma(\rho_\chi + \rho + 2)} \sum_{k \leq N-1} \Delta_{1/N}^2\left(f_{Zyg}, \frac{k}{N}\right) k^{\rho + \rho_\chi + 1} \\ & \quad + O_{f_{Zyg}, \varepsilon, \theta} \left(N^{3/2} \log(qT)^2 + \frac{N^2 \log(qNT)^2}{T} \right. \\ & \quad \left. + \log(q)^3 N^{3/2 + \varepsilon} \log(N)^3 \left[1 + \frac{N \log(qT)^2}{T} \right] \left[1 + \frac{N^{1-\varepsilon}}{U^{1-\varepsilon}} \right] \right). \end{aligned}$$

From the previous theorem, we derive the following upper bound.

Theorem 32. *Assume all the hypotheses of the previous theorem and fix $0 < \varepsilon < 1/4$ and $\theta = 1 - 2\varepsilon$. We get*

$$\sum_{\substack{1 < q \leq N^{1/2-2\varepsilon} \\ (N, q) = 1}} \frac{1}{\varphi(q)} \left| \sum_{\chi \neq \chi_0} \bar{\chi}(N) \sum_{n \leq N} \sum_{m \leq N-n} \Lambda(n) \chi(n) \mu(m) f_{Zyg}\left(\frac{n+m}{N}\right) \right| \ll_{f_{Zyg}, \varepsilon} N^{2-\varepsilon}.$$

ACKNOWLEDGEMENTS

The author is member of the Gruppo Nazionale per l'Analisi Matematica, la Probabilità e le loro Applicazioni (GNAMPA) of the Istituto Nazionale di Alta Matematica (INdAM). The author thanks Professor Alessandro Zaccagnini and Professor Alessandro Gambini for some discussions on this topic.

Conflict of interest. The author declare that he has no conflict of interest.

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